

Note on canonical quantization of Chern-Simons theory

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Abstract

This note discusses the canonical quantization of the Chern-Simons theory. We will focus on the case where the spatial geometry is a disk. The discussion will be helpful in understanding the entanglement entropy in Chern-Simons theory. [1–6]

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1 Introduction

In the zoo of quantum field theories (QFTs), a simple and interesting class is the topological quantum field theories (TQFTs). In such kind of theories, all observables are invariant under smooth deformation of the spacetime manifold, or equivalently, independent of the choice of spacetime coordinate.¹ For example, under a transformation $x \rightarrow x' = f(x)$, the correlation function of scalar operators must satisfy (we ignore the boundary)

$$\langle \phi(x_1)\phi(x_2)\cdots \rangle = \langle \phi(x'_1)\phi(x'_2)\cdots \rangle \quad (1.1)$$

It implies that $\phi \equiv c$ is just a c-number. More generally, one can show that there are no local operators in the theory. Therefore, TQFTs only have a finite number of degrees of freedom and hence a finite dimensional Hilbert space (if we ignore the boundary).

One important example is the three-dimensional Chern-Simons gauge theory, which is the focus of this note. Let Y denote the space-time manifold and A the gauge field, the action takes the following form

$$S[A] = \frac{k}{4\pi} \int_Y \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \quad (1.2)$$

where k is the coupling constant and is called the level of the Chern-Simons theory. It was first studied in details in [1, 2] and then plays important roles in high energy, condensed matter physics and mathematical physics. In this note, we will explain its canonical quantization. The detail of the quantization procedure depends on the choice of spatial manifold. Here we primarily work on a spatial unit disk, i.e., $Y = D^2 \times R$. There are two main reasons for studying this choice:

- It exhibits a clear connection to the (1+1)-dimensional conformal field theories. We will see that the physical degrees of freedom are only on the *edge*, which are described by a chiral Wess-Zumino-Witten model. The Hilbert space forms a representation of a chiral Kac-Moody algebra, the unitarity of which imposes quantization condition on the level k . The details of the dynamics (such as the energy) depends on the choice of the boundary condition.
- It is relevant to defining the bipartite entanglement in Chern-Simons theory. The definition of entanglement requires a factorization of the Hilbert space as the prerequisite. For the bipartite entanglement, we need

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \quad (1.3)$$

In the context of QFTs, A and B will be a spatial region and its complement. However, the definition of $\mathcal{H}_A$ and $\mathcal{H}_B$ needs boundary conditions, which are not part of the data in the original QFT. Therefore, the factorization (1.3) does not make sense (and hence, does not exist).²

¹The value of the correlation function is always independent of the choice of coordinate. Here, we require the form, the way it depends on the coordinate, is also invariant.

²A more intrinsic statement on the fact that any QFT does not have a factorized Hilbert space is that the local algebra of QFTs is a type III von Neumann factor. We thank Xi Yin for this comment.

One solution of this issue is the cutting map method. The idea is to first add appropriate boundary conditions at the entangling surface to define a Hilbert space for each region and get a factorized Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$. Then one embeds the original Hilbert space into this factorized one through an isometry

$$i_\epsilon : \mathcal{H} \rightarrow \mathcal{H}_A \otimes \mathcal{H}_B \quad (1.4)$$

to define various entanglement quantities therein. Thus, one has to understand how to define Hilbert space of the QFT on spatial manifolds with boundary to actually implement this method. The disk the simplest case.

The rest of the note is organized as follows. In Sec. 2, we introduce the non-Abelian Chern-Simons action and its basic properties. In Sec. 3, we discuss the canonical quantization on a disk in details. The appendix contains some useful details and other related topics. Appendix. B review the canonical formalism of constrained systems and the quantization. The main text focuses on the non-abelian Chern-Simons theory. Appendix. C reviews the Abelian version, which is also quite useful and has simplified features.

2 Basic properties of the Chern-Simons theory

We consider the Chern-Simons gauge theory of a simply connected Lie group G . Most of our statement holds more generally though. After introducing the action, we will examine its symmetries and equations of motion.

Action Let $A = A_\mu^a T^a dx^\mu$ be the gauge field that takes the value in the Lie algebra. In this convention, the generators of the Lie algebra T^a are anti-Hermitian and satisfy

$$[T^a, T^b] = f^{ab}_c T^c, \quad \text{Tr } T^a T^b = -\delta^{ab} \quad (2.1)$$

The gauge field transforms in the following way

$$A \mapsto A^g \equiv g A g^{-1} - dg g^{-1}, \quad g \in \text{SU}(N) \quad (2.2)$$

so that $(A^g)^h = A^{hg}$. Note that g is unitary, thus $dg g^{-1}$ is anti-Hermitian as well. The field strength is defined as

$$F = \frac{1}{2} F_{\mu\nu} dx_\mu \wedge dx_\nu = dA + A \wedge A, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad (2.3)$$

The theory is defined by a classical action

$$\begin{aligned} S[A] &= \frac{k}{4\pi} \int_Y \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \\ &= \frac{k}{4\pi} \int_Y dx^3 \epsilon^{\mu\nu\rho} \text{Tr} \left(A_\mu \partial_\nu A_\rho + \frac{2}{3} A_\mu A_\nu A_\rho \right) \end{aligned} \quad (2.4)$$

where Y is a space-time manifold and k is the coupling constant, called the level. We denote the above Chern-Simons theory by G_k .

Time reversal and parity The presence of $\epsilon^{\mu\nu\rho}$ implies that the action changes sign under the parity or time-reversal transformation. This is one of the reasons why the Chern-Simons theory can describe the fractional quantum Hall systems. This is also consistent with the fact that putting the system on a disk gives rise to chiral edge modes, which will be shown later.

Let us consider the time-reversal in more details. There is always ambiguity in the definition of time reversal, here, we work with the following one³

$$\mathcal{T} : A_0(t) \mapsto A'_0(t) = -A_0(-t), A_i(t) \mapsto A'_i(t) = A_i(-t) \quad (2.5)$$

Let us use the rule to compute how the the action transforms under the time-reversal

$$S'[A'] = \frac{k}{4\pi} \int dt' d^2x \epsilon^{\mu\nu\rho} \text{Tr} \left(A'_\mu(t', x) \partial'_\nu A'_\rho(t', x) + \frac{2}{3} A'_\mu(t', x) A'_\nu(t', x) A'_\rho(t', x) \right)$$

where the derivative is $\partial'_\mu = (\partial_{t'}, \partial_x) = (-\partial_t, \partial_x)$. The first term contains derivatives that transform in the same way as the gauge field, and thus changes the sign, e.g.,

$$\int dt' d^2x \text{Tr} A_i(t') \partial_{t'} A_j(t') = \int dt d^2x \text{Tr} A'_i(-t) \partial_{-t} A'_j(-t) = - \int dt d^2x \text{Tr} A_i(t) \partial_t A_j(t)$$

The second term does not involve derivative and always contains A_0 , thus must changes sign as well, i.e.,

$$\int dt' d^2x \epsilon^{\mu\nu\rho} \text{Tr} (A'_\mu(t', x) A'_\nu(t', x) A'_\rho(t', x)) = - \int dt d^2x \epsilon^{\mu\nu\rho} \text{Tr} (A_\mu(t, x) A_\nu(t, x) A_\rho(t, x))$$

Therefore, the action changes its sign under the time-reversal transformation

$$\mathcal{T} : S[A] \mapsto S'[A'] = -S[A] \quad (2.6)$$

The demonstration of the parity symmetry breaking is similar.

Gauge transformation As a gauge theory, the action has to be invariant under the gauge transformation. This turns out to be a subtle and interesting issue for the Chern-Simons theory. Let us explicitly compute $S[A^g]$. The first term in the action gives

$$\begin{aligned} & \text{Tr} ((gAg^{-1} - dgg^{-1}) \wedge d(gAg^{-1} - dgg^{-1})) \\ &= \text{Tr} ((gAg^{-1} - dgg^{-1}) \wedge (dgAg^{-1} + gdAg^{-1} - gAdg^{-1} + dgdg^{-1})) \\ &= \text{Tr} (AdA + 2A^2g^{-1}dg - 2Ag^{-1}dgg^{-1}dg + d(g^{-1}dgA) + g^{-1}dgg^{-1}dgg^{-1}dg) \end{aligned}$$

The second term in the action gives

$$\text{Tr} (A^3 - 3A^2g^{-1}dg + 3Ag^{-1}dgg^{-1}dg - g^{-1}dgg^{-1}dgg^{-1}dg)$$

³In our convention, the gauge field A_μ is anti-Hermitian (purely imaginary in the Abelian case), which explains why the transformation looks opposite to the conventional one. For example, the time reversal in the Maxwell theory is usually defined as $\mathcal{T} : E \mapsto E, B \mapsto -B$ or $\mathcal{T} : A_0(t) \mapsto A'_0(t) = A_0(-t), A_i(t) \mapsto A_i(t) = -A_i(-t)$, where A_μ is a real field.

In total, we have

$$S[A^g] - S[A] = \frac{k}{4\pi} \int_{\partial Y} g^{-1} dg A + \frac{k}{12\pi} \int_Y \text{Tr} (g^{-1} dg g^{-1} dg g^{-1} dg) \quad (2.7)$$

The first term is a boundary term. The second term is the Wess-Zumino (WZ) term denoted by Γ

$$S[A^g] = S[A] + k\Gamma[g], \quad \Gamma[g] = \frac{1}{12\pi} \int_Y \text{Tr} (g^{-1} dg g^{-1} dg g^{-1} dg) \quad (2.8)$$

where we choose the normalization such that $\Gamma[g]$ takes values in $2\pi\mathbb{Z}$. The WZ terms can have non-trivial effect because the space of maps $g : Y \rightarrow G$ is not simply connected. For example, suppose the space-time manifold is a three-sphere S^3 ,⁴ the gauge transformations are maps

$$g : S^3 \rightarrow G,$$

which are classified by $\pi_3(G)$. For any simply connected Lie group G , we have $\pi_3(G) = \mathbb{Z}$ so that the gauge transformations fall into sectors labeled by a "winding number", which is exactly captured by the WZ term. Under a transformation of winding number m , the Chern-Simons action transforms as

$$S[A^g] = S[A] + 2\pi km \quad (2.9)$$

In the path-integral quantization, we requires $e^{iS[A^g]}$ is invariant under the gauge transformation, which gives a restriction $k \in \mathbb{Z}$. In this note, we will perform the canonical quantization and get the same conclusion. Since the WZ term only detects the topological effect, it does not appear in the equation of motion (local dynamics).

Equations of motion The variation of the action

$$\begin{aligned} \delta S &= \frac{k}{4\pi} \int_Y \text{Tr} \left(\delta A \wedge dA + A \wedge d\delta A + \frac{2}{3} (\delta A \wedge A \wedge A + A \wedge \delta A \wedge A + A \wedge A \wedge \delta A) \right) \\ &= \frac{k}{4\pi} \int_Y \text{Tr} (2\delta A (dA + A \wedge A) + d(A \wedge \delta A)) \\ &= \frac{k}{4\pi} \int_{\partial Y} \text{Tr} (\delta A \wedge A) + \frac{k}{2\pi} \int_Y \text{Tr} (\delta A \wedge F) \end{aligned} \quad (2.10)$$

From the first to the second line, we used the cyclicity of the trace. From the second to the last line, we used the cyclicity of the trace and Stoke's theorem. Thus, in the absence of a boundary, the equations of motion is

$$F = 0 \quad (2.11)$$

Namely, there is no local physical degrees of freedom. We will see below that the presence of a boundary will make the dynamics more interesting.

⁴If Y is non-compact, such as R^3 , the gauge transformation is required to approach the identity at the spatial and temporal infinity, i.e., $\lim_{x \rightarrow \infty} U(x) = \mathbb{I}$. This is equivalent to considering an S^3 .

3 Canonical quantization on a disk

In this section, we explain how to quantize the Chern-Simons theory on a disk. First, we discuss how to properly describe the classical theory in the canonical formalism, i.e., construct the phase space and Poisson structure. Then, quantization is simply replacing the Poisson (or Dirac) bracket with the quantum commutator, which, as we will see, is the Kac-Moody algebra. Because the Hilbert space has to be a unitary representation, this will impose constraints on the coupling constant and the allowed states.

3.1 Effective action for the boundary

As we have seen above, the Chern-Simons theory on a closed spatial manifold does not have local physical degrees of freedom. The presence of a boundary gives rise new dynamical degrees of freedom that are chiral and only live on the boundary. In this subsection, we explain how the boundary affects the dynamics and derive the action for the boundary modes.

First of all, we need to determine the boundary condition. A general criterion is to choose the boundary conditions such that there are no boundary contribution to the equations of motion, i.e.,

$$\frac{k}{4\pi} \int_{\partial Y} \text{Tr}(\delta A \wedge A) = 0 \quad (3.1)$$

Because we consider the manifold $Y = D^2 \times \mathbb{R}$ with $\mathbb{R}$ being the time direction, a convenient choice is⁵

$$A_0|_{\partial Y} = 0 \quad (3.4)$$

Therefore, a classical Chern-Simons theory on $Y = D^2 \times R$ should be defined as follows

$$S[A] = \frac{k}{4\pi} \int \text{Tr} \left[AdA + \frac{2}{3} A^3 \right], \quad A_0|_{\partial Y} = 0 \quad (3.5)$$

The boundary condition also disallows some gauge transformations. Now, the legitimate gauge transformations are ones that are independent of time on the boundary and thus do not change the boundary conditions. However, only a subgroup of this group should be viewed as a gauge symmetry. This is the set of transformations that are one on the boundary, i.e.,

$$g|_{\partial Y} = 1 \quad (3.6)$$

⁵In order to for the gauge field to have a non-zero boundary value, one needs to add a boundary term. For example, by adding

$$\frac{k}{4\pi} \int_{\partial Y} \text{Tr} A_t A_\sigma \quad (3.2)$$

to the action, the boundary term in the variation of the action becomes

$$\frac{k}{4\pi} \int \text{Tr}(\delta A_t A_\sigma - \delta A_\sigma A_t) + \text{Tr}(\delta A_t A_\sigma + A_t \delta A_\sigma) = \frac{k}{2\pi} \int \text{Tr} \delta A_t A_\sigma \quad (3.3)$$

Thus, it is legitimate to have (fixed) nonzero values of A_t at the boundary while maintaining a well-defined variational problem. This trick is useful when one wants to use the path integral to obtain the wavefunction.

Time independent transformation on the boundary should be viewed as a *global symmetry*, because they does not approach identity in the temporal infinity. These are

$$g\Big|_{\partial Y} = g(\sigma) \quad (3.7)$$

where $\sigma \in [0, 2\pi]$ parametrizes the boundary circle. These transformations form a *loop group*, the elements of which are loops in a Lie group. The spectrum should be in the representation of these symmetries rather than being invariant under them.

In order to find its Hamiltonian formalism, it is more convenient to isolate the time direction and identify the dynamical variables. We decompose the field A and derivatives d into spatial and temporal components and write the action as

$$S[A] = -\frac{k}{4\pi} \int \text{Tr} \tilde{A} \partial_t \tilde{A} + \frac{k}{2\pi} \int \text{Tr} [A_0(\tilde{d}\tilde{A} + \tilde{A}^2)] \quad (3.8)$$

where $\tilde{A}, \tilde{d}$ denote the spatial components. Note that A_0 does not have dynamics and implements the Gauss law constraints on the spatial components

$$\tilde{F} = \tilde{d}\tilde{A} + \tilde{A}^2 = 0 \quad (3.9)$$

Thus, we can remove A_0 by choosing the temporal gauge $A_0 = 0$, which is the most convenient choice that is compatible with the boundary condition, *and* impose the Gauss law. The Gauss law constraint can be satisfied automatically if we introduce a new variable $U : D^2 \times R \rightarrow G$ and define

$$\tilde{A} = -\tilde{d}U U^{-1} \quad (3.10)$$

Notice that this reparametrization has a gauge redundancy, namely, $U \mapsto UV(t)$ leaves $\tilde{A}$ invariant. Now we substitute $\tilde{A} = -\tilde{d}U U^{-1}$ into the gauge-fixed action and have

$$\begin{aligned} S &= -\frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} A_i \partial_t A_j \quad (i, j \text{ are spatial indices}) \\ &= -\frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} [\partial_i U U^{-1} \partial_t (\partial_j U U^{-1})] \\ &= -\frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} [\partial_i U U^{-1} (\partial_t \partial_j U) U^{-1}] - \frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} [\partial_i U U^{-1} \partial_j U \partial_t U^{-1}] \\ &= -\frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} [\partial_i U^{-1} (\partial_t \partial_j U) U^{-1}] + \frac{k}{4\pi} \int \epsilon^{ij} \text{Tr} [U^{-1} \partial_t U U^{-1} \partial_i U U^{-1} \partial_j U] \end{aligned}$$

where the second term is proportional to the WZ term. Note that

$$\begin{aligned} \epsilon^{ij} \partial_j \text{Tr} [\partial_i U^{-1} \partial_t U] &= \epsilon^{ij} \text{Tr} [\partial_i U^{-1} \partial_t \partial_j U] \\ &= -\epsilon^{ij} \text{Tr} [U^{-1} \partial_i U U^{-1} (\partial_t \partial_j U)] \end{aligned}$$

which is exactly the first term above. Thus, we have

$$S[U] = \frac{k}{4\pi} \int_{\partial Y} \text{Tr} \partial_\sigma U^{-1} \partial_t U + \frac{k}{12\pi} \int_Y \text{Tr} (U^{-1} dU)^3 \quad (3.11)$$

where σ is the coordinate for the boundary S^1 . This is the chiral version of the Wess-Zumino-Witten model and the coefficients of the two terms happen to be right at the critical point.⁶

Remark The effective action for the boundary does not have Lorentz invariance. This is related to the boundary condition that we choose. A more general boundary condition is the following

$$A_0 - vA_\sigma \Big|_{\partial Y} = 0 \quad (3.13)$$

where v is an arbitrary constant. Then, one has to use a different gauge fixing that is compatible with this boundary condition. It can be shown effective action for the boundary becomes

$$S[U] = \frac{k}{4\pi} \int_{\partial Y} \text{Tr} \partial_\sigma U^{-1} (\partial_t U - v^2 \partial_\sigma U) + k\Gamma[U] \quad (3.14)$$

When $v = 1$, this yields a Lorentz invariant theory.

Let us derive the equation of motion. The variation of the WZ term is a surface term. The variation of the chiral kinetic term is

$$\begin{aligned} \delta S_0 &= \frac{k}{4\pi} \int_{\partial Y} \text{Tr} (\partial_\sigma \delta U^{-1} \partial_t U + \partial_\sigma U^{-1} \partial_t \delta U) \\ &= \frac{k}{4\pi} \int_{\partial Y} \text{Tr} (-\delta U^{-1} \partial_\sigma \partial_t U - (\partial_\sigma \partial_t U^{-1}) \delta U) \quad (\text{integration by part}) \\ &= \frac{k}{4\pi} \int_{\partial Y} \text{Tr} (U^{-1} \delta U (2U^{-1} \partial_\sigma \partial_t U - U^{-1} \partial_t U U^{-1} \partial_\sigma U - U^{-1} \partial_\sigma U U^{-1} \partial_t U)) \end{aligned}$$

Combined together, we have

$$\begin{aligned} \delta S[U] &= \frac{k}{2\pi} \int_{\partial Y} \text{Tr} (U^{-1} \delta U (U^{-1} \partial_\sigma \partial_t U - U^{-1} \partial_\sigma U U^{-1} \partial_t U)) \\ &= \frac{k}{2\pi} \int_{\partial Y} \text{Tr} U^{-1} \delta U \partial_\sigma (U^{-1} \partial_t U) \end{aligned} \quad (3.15)$$

If the coefficients of the kinetic term and the WZ term are random numbers, the variation of the action will be more complicated. Thus, the equation of motion is

$$\partial_\sigma (U^{-1} \partial_t U) = 0 \quad (3.16)$$

⁶The standard WZW model is a principal chiral field action (written with the holomorphic and anti-holomorphic derivatives) plus a WZ term

$$S_{WZW}[g] = \frac{1}{\lambda^2} \int \text{Tr} [\partial_z g^{-1} \partial_{\bar{z}} g] + k\Gamma[g], \quad k \in \mathbb{Z} \quad (3.12)$$

It is a conformal field theory only when the coefficients satisfies $\lambda^2 = 4\pi/k$.

3.2 Poisson (Dirac) bracket

After identifying the dynamical variables as the chiral boundary modes $U(\sigma)$, we discuss how to construct the phase space and find the Poisson bracket. The chiral WZW action is first order in time derivative. The canonical formalism of such kind of system is complicated due to the fact that the canonical momenta is not independent of the canonical coordinates. Therefore, one has to define the phase space and the bracket that incorporate such constraints properly. A systematic way is due to Dirac and is reviewed in Appendix. B. However, the chiral WZW action is somewhat special and allows for shortcut. [7]

Before discussing the chiral WZW action, let us begin with a more general theory with dynamical variables ϕ^i and an arbitrary action that is first order in time derivative

$$I = \int \left(A_i(\phi) \frac{d\phi^i}{dt} - H(\phi) \right) dt \quad (3.17)$$

where i can be regarded as a collection of indices and even can be continuous (say, spatial coordinate in a field theory), $H(\phi)$ denotes terms independent of time derivatives. Such terms are important in the equations of motion but do not play any role in determining the canonical formalism and computing the brackets. We can use the least action principle to find the equations of motion as

$$\left(\frac{\partial A_i}{\partial \phi^j} - \frac{\partial A_j}{\partial \phi^i} \right) \frac{d\phi^i}{dt} = \frac{\partial H}{\partial \phi^j} \quad (3.18)$$

Now, we want to find the canonical formalism that correctly reproduces the dynamics. It can be done in two different but equivalent ways:

- *Dirac's approach.* We regard ϕ^i as the canonical coordinate, whose conjugate momentum is defined as

$$\pi_i = \frac{\partial L}{\partial \dot{\phi}^i} = A_i(\phi) \quad (3.19)$$

Thus, it gives a primary constraint

$$\chi_i = \pi_i - A_i(\phi) = 0 \quad (3.20)$$

We can define the Poisson bracket as usual on this phase space (ϕ, π)

$$\{\phi_i, \pi_j\}_P := \delta_{ij} \quad (3.21)$$

The Hamiltonian is just $H(\phi)$ that is independent of the canonical momentum. Let us assume that the above primary constraints are of the second class, i.e.,

$$F_{ij} = \{\chi_i, \chi_j\}_P = \frac{\partial \chi_i}{\partial \phi^k} \frac{\partial \chi_j}{\partial \pi^k} - \frac{\partial \chi_i}{\partial \pi^k} \frac{\partial \chi_j}{\partial \phi^k} = \frac{\partial A_i}{\partial \phi^j} - \frac{\partial A_j}{\partial \phi^i} \quad (3.22)$$

is non-degenerate, also guarantees that there will be no more constraints being generated.⁷ We can follow the general recipe to define the Dirac bracket. The Dirac bracket between the canonical coordinates is no longer zero. For any two functions of ϕ , we have

$$\{X, Y\}_D = -\{X, \chi_i\}_P (F^{-1})_{ij} \{\chi_j, Y\} = F_{ij}^{-1} \frac{\partial X}{\partial \phi^i} \frac{\partial Y}{\partial \phi^j} \quad (3.23)$$

⁷Namely, the equation $[\chi_i, H] + u_j [\chi_i, \chi_j] = 0$ cannot be reduced to an equation that is independent of u .

- *Witten's approach.* Since the physical quantities are eventually functions of ϕ 's only, it seems to be redundant to introduce the canonical momentum. Here, we let the phase space contain only ϕ and find a Poisson bracket on it. Namely, we want to find a non-degenerate 2-form $w = \frac{1}{2}w_{ij}d\phi^i \wedge d\phi^j$ so that the Poisson bracket between two functions X, Y on the phase space is

$$\{X, Y\}_P := w_{ij} \frac{\partial X}{\partial \phi^i} \frac{\partial Y}{\partial \phi^j} \quad (3.24)$$

Moreover, we require the Poisson bracket gives the correct equations of motion, i.e.,

$$\frac{dX}{dt} = \{X, H(\phi)\}_P \quad (3.25)$$

The variation of the action is

$$\begin{aligned} \delta I &= \int \left(\frac{\partial A_i}{\partial \phi^j} - \frac{\partial A_j}{\partial \phi^i} \right) \delta \phi^j \frac{d\phi^i}{dt} - \frac{\partial H}{\partial \phi^j} \delta \phi^j \\ &= \int F_{ij} \delta \phi^j \frac{d\phi^i}{dt} - \frac{\partial H}{\partial \phi^j} \delta \phi^j \end{aligned} \quad (3.26)$$

Notice that F_{ij} , as the coefficient of $\delta \phi^i \frac{d\phi^j}{dt}$, is always anti-symmetric and can naturally induce a symplectic structure. We can define the Poisson bracket of any two functions on the phase space to be

$$\{X, Y\}_P = F_{ij}^{-1} \frac{\partial X}{\partial \phi^i} \frac{\partial Y}{\partial \phi^j} \quad (3.27)$$

This choice can gives the correct equations of motion and is the same as the result (3.23) obtained in the first approach, as it has to be.

Let us make a few comments on the two approaches:

- Although the two equations (3.23) and (3.27) take the same form, they are defined on different phase spaces. The former includes the momentum and is twice as large as the latter one.
- The first approach is more systematic and can be applied to any systems. Generalizing the second approach amounts to writing the action in a first-order form on the phase space.⁸ In the most general situations (with constraints), writing the action in the phase space is as hard as the first approach. However, when the action is already in the first order time derivative, this approach is much more convenient than the first one.
- In the first approach, one has to choose an explicit set of coordinates ϕ^i in order to compute the conjugate momenta. In the second approach, it is enough to have a basis of tangent vectors to the phase space and the matrix F_{ij} may be constructed relative to any such basis. This is particularly convenient when the configuration is nonlinear.

⁸In the most standard case, we have

$$\delta S = \delta \int (\pi_q \dot{q} - H(\pi, q)) dt \quad (3.28)$$

Its variation yields the standard Poisson bracket.

The chiral WZW model is a non-linear sigma model. It is awkward to explicitly work with any coordinate system. Thus, we will use the second approach to find the canonical formalism.⁹ As we have commented above, we only need to find *a basis* of tangent vectors. Here, a very convenient basis of tangent vectors are the matrices $U^{-1}\delta U$ (in contrast, δU would be an inconvenient choice). The matrix F then must act both on the Lie algebra index of $U^{-1}\delta U$ and the spatial coordinate σ . Recall the variation of the action

$$\delta S = \frac{k}{2\pi} \int_{\partial Y} \text{Tr} U^{-1} \delta U \partial_\sigma (U^{-1} \partial_t U) \quad (3.29)$$

Thus, we see that

$$F = \mathbb{I} \otimes \frac{n}{2\pi} \partial_\sigma, \quad (3.30)$$

where $\mathbb{I}$ acts on the Lie algebra index and ∂_σ acts on the spatial coordinate, whose inverse is $\theta(\sigma) + \text{const}$ ($\theta(\sigma)$ is the step function).

We now apply the definition of the Poisson bracket to

$$X(\sigma) = \text{Tr} A \frac{dU}{d\sigma} U^{-1}, \quad Y(\sigma') = \text{Tr} B \frac{dU}{d\sigma'} U^{-1}$$

where A and B are just auxiliary matrices that contract with the Lie algebra elements. The way to compute their Poisson bracket is to calculate $\delta X \delta Y = \frac{\partial X}{\partial \phi^i} \frac{\partial Y}{\partial \phi^j} \delta \phi^i \delta \phi^j$ and then replace $\delta \phi^i \delta \phi^j$ by F_{ij}^{-1} . First, we calculate

$$\begin{aligned} \delta X &= \text{Tr} A \left(\frac{d}{d\sigma} \delta U \right) U^{-1} + \text{Tr} A \frac{dU}{d\sigma} \delta U^{-1} \\ &= \text{Tr} A \left(\frac{d}{d\sigma} \delta U \right) U^{-1} - \text{Tr} A \frac{dU}{d\sigma} U^{-1} \delta U U^{-1} \\ &= \text{Tr} U^{-1} A U \frac{d}{d\sigma} (U^{-1} \delta U) \end{aligned}$$

so we have

$$\delta X(\sigma) \delta Y(\sigma') = \text{Tr} U^{-1} A U \frac{d}{d\sigma} (U^{-1} \delta U) \times \text{Tr} U^{-1} B U \frac{d}{d\sigma'} (U^{-1} \delta U)$$

By substituting $\delta \phi^i \delta \phi^j$ with F^{ij} , we have

$$\begin{aligned} \{X(\sigma), Y(\sigma')\}_P &= -\frac{2\pi}{k} \delta'(\sigma - \sigma') \text{Tr} U^{-1}(\sigma) A U(\sigma) U^{-1}(\sigma') B U(\sigma') \\ &= -\frac{2\pi}{k} \delta(\sigma - \sigma') \text{Tr} \left([A, B] \frac{dU}{d\sigma} U^{-1} \right) - \frac{2\pi}{k} \delta'(\sigma - \sigma') \text{Tr}(AB) \end{aligned} \quad (3.31)$$

A particular useful field is the *current* field

$$J^a = \frac{k}{2\pi} \text{Tr} T^a \frac{dU}{d\sigma} U^{-1} \quad (3.32)$$

⁹If we consider Abelian Chern-Simons theories, the boundary modes are described by chiral compact boson. Then it is not hard to compute the Poisson (Dirac) bracket in the first approach. We leave this discussion to Appendix. C.

where the factor of $\frac{k}{2\pi}$ is added to be consistent with the standard normalization. One can check that J^a is a real field, i.e.,

$$(J^a(\sigma))^* = -\frac{k}{2\pi} \text{Tr} U \frac{dU^{-1}}{d\sigma} T^a = \frac{k}{2\pi} \text{Tr} \frac{dU}{d\sigma} U^{-1} T^a = J^a(\sigma) \quad (3.33)$$

By using $[T^a, T^b] = f^{abc} T^c$ and $\text{Tr}[T^a T^b] = -\delta^{ab}$, we find

$$\begin{aligned} \{J^a(\sigma), J^b(\sigma')\}_P &= -\frac{k}{2\pi} \delta(\sigma - \sigma') \text{Tr} \left([T^a, T^b] \frac{dU}{d\sigma} U^{-1} \right) - \frac{k}{2\pi} \delta'(\sigma - \sigma') \text{Tr}(T^a T^b) \\ &= \delta(\sigma - \sigma') f^{abc} J^c(\sigma) + \frac{k}{2\pi} \delta'(\sigma - \sigma') \delta^{ab} \end{aligned} \quad (3.34)$$

Equivalently, we can write it in the momentum space. Introducing the Fourier modes,

$$J_n^a = \int d\sigma e^{in\sigma} J_a(\sigma), \quad (J_n^a)^* = J_{-n}^a \quad (3.35)$$

we can write the Poisson bracket in a nicer form

$$\{J_n^a, J_m^b\}_P = f^{abc} J_{n+m}^c - ikn \delta_{ab} \delta_{n+m,0} \quad (3.36)$$

If we trace back to how we reduce the gauge field A to the boundary degrees of freedom, we can find that variable J^a is related to the boundary value of A_σ as (recall $A_\sigma = A_\sigma^a T^a$)

$$J^a(\sigma) = \frac{k}{2\pi} A_\sigma^a(\sigma) \Big|_{\partial Y} \quad (3.37)$$

And the Fourier zero mode is the holonomy of A

$$J_0^a = \frac{k}{2\pi} \int d\sigma A_\sigma^a \Big|_{\partial Y} \quad (3.38)$$

3.3 Quantization, Kac-Moody algebra and its Representation

The canonical quantization is readily done. We only have to promote the Poisson bracket (3.36) into a quantum commutator in the standard way, i.e.,

$$[J_n^a, J_m^b] = i f^{abc} J_{n+m}^c + kn \delta_{ab} \delta_{n+m,0} \quad (3.39)$$

where J_n^a are now operators acting on a Hilbert space. Eq. (3.39) is known as the *Kac-Moody algebra*, which a prototype example of infinite dimensional Lie algebra. The Hilbert space should form a unitary representation of it. The unitarity is implemented as the Hermiticity of the generators

$$J^{a\dagger}(\sigma) = J^a(\sigma) \Leftrightarrow J_n^{a\dagger} = J_{-n}^a \quad (3.40)$$

and having a positive-definite norm. In the following, we will show that the unitarity of the representation (1) requires k to be an integer (2) restricts the allowed representations. For simplicity, we consider the simplest example $G = SU(2)$. The arguments below are readily

generalized to other simply connected Lie groups by examining the $SU(2)$ subgroup (and the corresponding subalgebra).

The irreducible representations of the Kac-Moody algebra are the highest weight representations. Namely, there are so-called highest weight states $|\lambda\rangle$ that transform as an irreducible representation of the ordinary Lie algebra of zero modes J_0^a , and are annihilated by the positive modes

$$J_n^a |\lambda\rangle = 0, \quad \forall n > 0 \quad (3.41)$$

New states can be generated by applying the negative modes J_{-n}^a to the highest weight states. Thus, the highest weight states $|\lambda\rangle$ are also called the *primaries* (of the Kac-Moody algebra) and states like $J_{-n}^a |\lambda\rangle$ are called the *descendants*.

Just like how we analyze an ordinary Lie algebra, it is convenient to use the Cartan basis. In a Cartan basis, the $J^a(z)$ are written as H^i and $E^{\pm\alpha}$, where $i = 1, \dots, r_G$ labels elements in the Cartan subalgebra and the positive roots α labels the raising and lowering operators. The highest weight states satisfy

$$\begin{aligned} H_0^i |\lambda\rangle &= \lambda^i |\lambda\rangle, \quad E_0^\alpha |\lambda\rangle = 0, \quad \alpha > 0 \\ H_n^i |\lambda\rangle &= E_n^\alpha |\lambda\rangle = 0, \quad n > 0 \end{aligned} \quad (3.42)$$

In our concrete example $G = SU(2)$,¹⁰ the the following linear combination

$$I^\pm = \frac{1}{\sqrt{2}}(J_0^1 \pm iJ_0^2), \quad I^3 = \frac{1}{\sqrt{2}}J_0^3 \quad (3.43)$$

gives a conventionally normalized $\mathfrak{su}(2)$ algebra $[I^+, I^-] = 2I^3, [I^3, I^\pm] = \pm I^\pm$, where $2I^3$ has integer eigenvalues in any finite dimensional representation. From the Kac-Moody algebra, we also find that

$$\tilde{I}^+ = \frac{1}{\sqrt{2}}(J_{+1}^1 - iJ_{+1}^2), \quad \tilde{I}^- = \frac{1}{\sqrt{2}}(J_{-1}^1 + iJ_{-1}^2), \quad \tilde{I}^3 = \frac{k}{2} - \frac{1}{\sqrt{2}}J_0^3 \quad (3.44)$$

also satisfy the standard $\mathfrak{su}(2)$ algebra $[\tilde{I}^+, \tilde{I}^-] = 2\tilde{I}^3, [\tilde{I}^3, \tilde{I}^\pm] = \pm\tilde{I}^\pm$ where $2\tilde{I}^3$ has integer eigenvalues. Thus k has to be an integer for unitary highest weight representation.¹¹

For a given level k , unitarity also puts constraints on what irreducible representation of the corresponding ordinary Lie algebra can appear. For example, we take $SU(2)_k$ and suppose there exists a highest weight representation in the spin- j representation of $SU(2)$. The $2j + 1$

¹⁰Notice the normalization $\text{Tr}[T^a T^b] = -\delta^{ab}$, we have $T^a = -\frac{i}{\sqrt{2}}\sigma^a$ (σ^a are the Pauli matrices) and $f^{abc} = \sqrt{2}\epsilon^{abc}$.

¹¹This argument can be generalized to other Lie algebra by using the canonical $\mathfrak{su}(2)$ subalgebra

$$I^\pm = E_0^{\pm\psi} \quad I^3 = \psi \cdot H_0 / \psi^2 \quad (3.45)$$

generated by the highest root ψ . The Kac-Moody algebra implies the following choice also form an $\mathfrak{su}(2)$ subalgebra.

$$\tilde{I}^\pm = E_{\pm 1}^{\mp\psi} \quad \tilde{I}^3 = \frac{k}{2} - \psi \cdot H_0 / \psi^2 \quad (3.46)$$

Thus, the level has to be quantized in a unitary representation.

states will be labeled by their I^3 eigenvalues, where I^3 is defined above. Using the other $\mathfrak{su}(2)$ generators, we can derive the most stringent condition by considering $|j, j\rangle$

$$0 \leq \langle j, j | \tilde{I}^+ \tilde{I}^- | j, j \rangle = \langle j, j | [\tilde{I}^+, \tilde{I}^-] | j, j \rangle = \langle j, j | k - 2I^3 | j, j \rangle = k - 2j \quad (3.47)$$

Therefore, unitarity only allows highest weight representation with¹²

$$2j \leq k \quad (3.49)$$

For example, in the case of $SU(2)_1$, there are only two highest weight representation with spin-0 and spin- $\frac{1}{2}$.

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A More on the Wess-Zumino term

In this section, we discuss some technical details of the Wess-Zumino term.

Abstract form First, let us write the WZ term in an abstract form. On the group manifold of any simple, non-abelian group G , there is a $G \times G$ invariant third rank tensor field w that is closed

$$dw = 0 \quad (A.1)$$

and locally but not globally exact

$$w = d\lambda \quad (A.2)$$

where λ is some second rank tensor field. We choose the normalization such that the integral of w over any three sphere in G is an integral multiple of 2π . Let Y be a three-dimensional manifold¹³ whose boundary ∂Y is identified with space-time. Given a mapping

$$g : \partial Y \rightarrow G$$

we can extend it to a mapping (also denoted by g) from Y to G . (The existence of this extension requires the map $g : \partial Y \rightarrow G$ is homotopic to a trivial map) The WZ term is defined as

$$\Gamma = \int_Y g^*(w) \quad (A.3)$$

¹²The above argument is also generalizable to other groups where one find

$$2\psi \cdot \lambda / \psi^2 \leq k \quad (3.48)$$

¹³In the standard definition, we choose a three-ball.

Here g^* is the pull-back of differential forms. Noting that $w = d\lambda$, we have

$$\Gamma = \int_Y g^*(d\lambda) = \int_{\partial Y} g^*(\lambda) \quad (\text{A.4})$$

The last formula exhibits Γ as the integral of an ordinary two-dimensional Lagrangian. In more concrete terms, let ϕ^i be a set of coordinates for the group manifold G and λ_{ij} be the components of the tensor λ . Then, the mapping $g : \partial Y \rightarrow G$ can be described by means of functions $\phi^i(x)$, where x is the coordinate on ∂Y and $\phi^i(x)$ gives the coordinate of the corresponding group element. The WZ term can be written as

$$\Gamma = \int_{\partial Y} g^*(\lambda_{ij} d\phi^i \wedge d\phi^j) = \int d^2x \epsilon^{\mu\nu} \lambda_{ij}(\phi(x)) \partial_\mu \phi^i(x) \partial_\nu \phi^j(x) \quad (\text{A.5})$$

This equation has Dirac string type singularities, because the defining equation of λ , $w = d\lambda$, can be solved only locally.

Explicit form for SU(2) Now, let us construct explicit formulas for the simplest non-abelian group SU(2). There are two commonly used parametrizations and we will discuss both of them.

Polar angle parametrization. The SU(2) manifold is a three sphere, which can be described by

$$x_0^2 + x_1^2 + x_2^2 + x_3^2 = 1 \quad (\text{A.6})$$

or polar angles $\psi \in [0, \pi]$, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$. The two set of coordinates are related by

$$x_0 = \cos \psi, \quad x_3 = \sin \psi \cos \theta, \quad x_1 = \sin \psi \sin \theta \cos \phi, \quad x_2 = \sin \psi \sin \theta \sin \phi \quad (\text{A.7})$$

The upper hemisphere $\psi \in [0, \frac{\pi}{2}]$ is a three-ball, which can be viewed as a foliation of two-spheres. The radius of the two-spheres are given by $\sin \psi$. A generic element in SU(2) is written as

$$g = x_0 + i \sum_{k=1}^3 x_k \sigma^k = \begin{pmatrix} x_0 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & x_0 - ix_3 \end{pmatrix} \quad (\text{A.8})$$

The line element is

$$ds^2 = -\frac{1}{2} \text{Tr} (g^{-1} dg g^{-1} dg) = d\psi^2 + \sin^2 \psi (d\theta^2 + \sin^2 \theta d\phi^2) \quad (\text{A.9})$$

which yields the volume form (the only third rank tensor in this case)

$$w = \frac{1}{\pi} \sin^2 \psi \sin \theta d\psi \wedge d\theta \wedge d\phi \quad (\text{A.10})$$

The normalization is chosen such that the integral of w over the whole manifold is 2π . The equation $w = d\lambda$ can be solved in many ways, for instance,

$$\lambda = \frac{1}{4\pi} (2\psi - \sin 2\psi) \sin \theta d\theta \wedge d\phi \quad (\text{A.11})$$

or

$$\lambda = \frac{1}{\pi} \phi \sin^2 \psi \sin \theta d\psi \wedge d\theta \quad (\text{A.12})$$

The WZ term is

$$\Gamma = \frac{1}{\pi} \int d^2x \phi(x) \sin^2 \psi(x) \sin \theta(x) \epsilon^{\mu\nu} \partial_\mu \psi \partial_\nu \theta \quad (\text{A.13})$$

Note that g is periodic in ϕ while the above expression of Γ is not! When ϕ changes by 2π , Γ changes by [i want to show the ambiguity is an integral multiple of 2π]

$$\begin{aligned} \Delta\Gamma &= 2 \int d^2x \sin^2 \psi(x) \sin \theta(x) \epsilon^{\mu\nu} \partial_\mu \psi \partial_\nu \theta \\ &= -2 \int d^2x \epsilon^{\mu\nu} \partial_\mu \left(\frac{\psi}{2} - \frac{\sin 2\psi}{4} \right) \partial_\nu (\cos \theta) \end{aligned} \quad (\text{A.14})$$

Euler angle parametrization. A different parametrization is to use the three Euler angles,

$$g = e^{i\phi \frac{\sigma^3}{2}} e^{i\theta \frac{\sigma^1}{2}} e^{i\psi \frac{\sigma^3}{2}}, \quad \phi \in [0, 2\pi], \theta \in [0, \pi], \psi \in [-2\pi, 2\pi] \quad (\text{A.15})$$

The line element is

$$ds^2 = \frac{1}{4} (d\phi^2 + d\theta^2 + d\psi^2 + 2 \cos \theta d\phi d\psi) \quad (\text{A.16})$$

which yields the following third rank tensor (normalized such that $\int_{SU(2)} w = 2\pi$)

$$w = \frac{1}{8\pi} \sin \theta d\phi \wedge d\theta \wedge d\psi \quad (\text{A.17})$$

The solution to λ can be chosen as

$$\lambda = \frac{1}{8\pi} \psi \sin \theta d\phi \wedge d\theta \quad (\text{A.18})$$

And the WZ term reads

$$\Gamma = \frac{1}{8\pi} \int d^2x \psi(x) \sin \theta(x) \epsilon^{\mu\nu} \partial_\mu \phi \partial_\nu \theta \quad (\text{A.19})$$

The ψ has a 4π ambiguity, which makes the WZ term also ambiguous

$$\Delta\Gamma = \frac{1}{2} \int d^2x \sin \theta(x) \partial_\mu \phi \partial_\nu \theta \quad (\text{A.20})$$

The integral is exactly how many times the two-dimensional manifold ∂Y covers the unit two-sphere with an area 4π , thus, $\Delta\Gamma \in 2\pi\mathbb{Z}$.

The Euler angle parametrization has another advantage compared with the polar angle. It has the following identities (which can be checked by using Mathematica)

$$\begin{aligned} -i \operatorname{Tr} (\sigma^3 g^{-1} dg) &= d\psi + \cos \theta d\phi \\ -i \operatorname{Tr} (\sigma^\pm g^{-1} dg) &= e^{\pm i\psi} (d\theta \mp i \sin \theta d\phi) \end{aligned} \quad (\text{A.21})$$

where $\sigma^\pm = \sigma^1 \pm i\sigma^2$. These expressions will be much more complicated in the polar angle parameterization. If we write

$$-ig^{-1}dg = \frac{1}{2} \sum_{k=1}^3 V_k \sigma^k = \frac{1}{2} \left(\frac{V_1 - iV_2}{2} \sigma^+ + \frac{V_1 + iV_2}{2} \sigma^- + V_3 \sigma^3 \right)$$

we have

$$V_1 \pm iV_2 = e^{\pm i\psi} (d\theta \mp i \sin \theta d\phi), \quad V_3 = d\psi + \cos \theta d\phi \quad (\text{A.22})$$

As a sanity check, one can compute

$$ds^2 = -\frac{1}{2} \text{Tr} (g^{-1} dg g^{-1} dg) = \frac{1}{4} ((V_1 - iV_2)(V_1 + iV_2) + V_3 V_3)$$

and find that it is the same as the line element we wrote above.

Variation

$$\begin{aligned} \delta\Gamma &= \frac{3}{12\pi} \int_Y \text{Tr} ((\delta U^{-1} dU + U^{-1} d\delta U) \wedge U^{-1} dU \wedge U^{-1} dU) \\ &= \frac{1}{4\pi} \int_Y \text{Tr} ((-U^{-1} \delta U U^{-1} dU + U^{-1} d\delta U) \wedge U^{-1} dU \wedge U^{-1} dU) \\ &= \frac{1}{4\pi} \int_Y d \text{Tr} ((U^{-1} \delta U)(U^{-1} dU) \wedge U^{-1} dU) \quad (\text{cyclicity, } d^2 = 0) \\ &= \frac{1}{4\pi} \int_{\partial Y} \text{Tr} ((U^{-1} \delta U)(U^{-1} dU) \wedge U^{-1} dU) \quad (\text{Stokes theorem}) \\ &= -\frac{1}{4\pi} \int_{\partial Y} \text{Tr} ((U^{-1} \delta U) d(U^{-1} dU)) \quad (d^2 = 0) \\ &= -\frac{1}{4\pi} \int_{\partial Y} \epsilon^{\mu\nu} \text{Tr} ((U^{-1} \delta U) \partial_\mu (U^{-1} \partial_\nu U)) \end{aligned} \quad (\text{A.23})$$

B Canonical quantization of constrained systems

B.1 Generalities of classical constrained systems

Our starting point is the action principle. We assume that there is an action integral which depends on the motion, such that, when one varies the motion and puts down the conditions for the action integral to be stationary, one gets the equations of motion. The method of starting from an action has the advantage that it makes the theory conform to the principle of relativity rather easily. The action integral will be denoted by

$$I = \int L dt \quad (\text{B.1})$$

One might wonder whether one could not take the Hamiltonian as the starting point and short-circuit this work of passing from the action integral to the Hamiltonian. The objection to trying to make this short-circuit is that it is not at all easy to formulate conditions for a theory to be relativistic in terms of the Hamiltonian. When we have the Hamiltonian, we can apply a standard method which gives us a first approximation to a quantum theory.

For simplicity, we start with a dynamical theory involving only a finite number of degrees of freedom. It is then merely a formal matter to pass from this to the infinite number of degrees of freedom which we need for a field theory.

Let q_n , $n = 1, \dots, N$, be the dynamical coordinates. Their velocities are $\dot{q}_n = dq_n/dt$. The Lagrangian is a function $L = L(q, \dot{q})$ of the coordinates and the velocities. The classical equations of motion are

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_n} \right) - \frac{\partial L}{\partial q_n} = 0 \quad (\text{B.2})$$

More explicitly, we have

$$\ddot{q}_m \frac{\partial^2 L}{\partial \dot{q}_m \partial \dot{q}_n} = \frac{\partial L}{\partial q_n} - \dot{q}_m \frac{\partial^2 L}{\partial q_m \partial \dot{q}_n} \quad (\text{B.3})$$

At a given instant of time, the state of the system is determined by the coordinates and velocities $(q, \dot{q})$. However, we see from Eq. (B.3) that the accelerations $\ddot{q}$ is uniquely determined if and only if the matrix $\frac{\partial^2 L}{\partial \dot{q}_m \partial \dot{q}_n}$ can be inverted, namely, if the determinant $\det(\frac{\partial^2 L}{\partial \dot{q}_m \partial \dot{q}_n})$ does not vanish. If, on the other hand, the determinant is zero, the acceleration is not uniquely determined and the solution of the equations of motion could then contain arbitrary functions of time that does not change the *physical state*.

The departing point for the Hamiltonian formalism is to define the canonical momenta

$$p_n := \frac{\partial L}{\partial \dot{q}_n} \quad (\text{B.4})$$

Here, we allow for the possibility of these momenta not being independent functions of the velocities. There may be several (independent) constraints

$$\phi_m(q, p) = 0, \quad m = 1, \dots, M \quad (\text{B.5})$$

They are called the primary constraints, a name due to Bergmann. The next step is to introduce the canonical Hamiltonian H . We start with the quantity $\dot{q}_n p_n - L$ and make variations in the variables q and $\dot{q}$

$$\begin{aligned} \delta(\dot{q}_n p_n - L) &= \delta \dot{q}_n p_n + \dot{q}_n \delta p_n - \delta \dot{q}_n \frac{\partial L}{\partial \dot{q}_n} - \delta q_n \frac{\partial L}{\partial q_n} \\ &= \dot{q}_n \delta p_n - \delta q_n \frac{\partial L}{\partial q_n} \end{aligned} \quad (\text{B.6})$$

We see that it does not involve $\delta \dot{q}$ explicitly but only through the combination $p(q, \dot{q})$. It means that $\dot{q}_n p_n - L$ can be expressed in terms of the q 's and p 's independent of the velocities, which is called the Hamiltonian H

$$H := \dot{q}_n p_n - L \quad (\text{B.7})$$

However, the Hamiltonian is not uniquely determined as a function of the q 's and p 's. Because δp are not all independent but are restricted to preserve the primary constraints $\phi_m \approx 0$. We may add to H any linear combination of ϕ_m 's, which are zero in the above variation. Thus, we could go over to another Hamiltonian

$$H^* = H + c_m(q, p) \phi_m(q, p) \quad (\text{B.8})$$

One can rewrite (B.6) as

$$\left(\frac{\partial H}{\partial q_n} + \frac{\partial L}{\partial q_n} \right) \delta q_n + \left(\frac{\partial H}{\partial p_n} - \dot{q}_n \right) \delta p_n = 0$$

where the variation δp_n and δq_n are subjected to the primary constraints, namely they are tangent to the constraint surfaces. Then one follow the general method of calculus and show that

$$\frac{\partial H}{\partial q_n} + \frac{\partial L}{\partial q_n} = -u_m \frac{\partial \phi_m}{\partial q_n}, \quad \frac{\partial H}{\partial p_n} - \dot{q}_n = -u_m \frac{\partial \phi_m}{\partial p_n}$$

where u_m is some unknown coefficient that can depend on p, q and the minus sign is added for convenience. In summary, the equivalent Hamiltonian form of the original Euler-Lagrangian equations is

$$\begin{aligned} \dot{q}_n &= \frac{\partial H}{\partial p_n} + u_m \frac{\partial \phi_m}{\partial p_n} \\ \dot{p}_n &= -\frac{\partial H}{\partial q_n} - u_m \frac{\partial \phi_m}{\partial q_n} \\ \phi_m(q, p) &= 0 \end{aligned} \tag{B.9}$$

where we have used the definition of canonical momenta.

It is convenient to introduce a certain formalism which enables one to write these equations in a more concise form, namely, the Poisson bracket formalism. The equations of motion can be written as

$$\dot{F} = [F, H] + u_m [F, \phi_m] \tag{B.10}$$

Let us now examine some of the consequences of the equations of motion. A basic consistency requirement is that the primary constraints should be preserved in time

$$\dot{\phi}_m = [\phi_m, H] + u_{m'} [\phi_m, \phi_{m'}] \equiv 0, \quad m = 1, \dots, M \tag{B.11}$$

It is possible for these consistency conditions to lead directly to an inconsistency (such as $1 = 0$). If that happens, it would mean that our original Lagrangian has self-inconsistency.¹⁴ After excluding this, (B.11) can be divided into three kinds:

- One kind reduces to $0 = 0$, i.e., it is identically satisfied with the help of the primary constraints.
- It reduces to an equation independent of the u 's, thus involving only the q and p . Such an equation must be independent of the primary constraints, which is called the secondary constraint

$$\chi(p, q) = 0 \tag{B.12}$$

Once a secondary constraint is found, we get yet another consistency condition. And we have to repeat the self-consistency check to exhaust all the possible constraints.

- It may impose a condition on the u 's.

$$[\phi_j, H] + u_m [\phi_j, \phi_m] = 0 \tag{B.13}$$

where $m = 1, \dots, M$ enumerates all the primary constraints and j takes on any of the primary or secondary constraints.

¹⁴One example is $L = q$, whose Euler-Lagrangian equation is $1 = 0$. Thus, we cannot take the Lagrangian to be completely arbitrary.

Let us look at these equations from the following point of view. Let us suppose that the u 's are unknowns and what we have here are a number of non-homogeneous linear equations in these unknowns u , with coefficients which are functions of the q 's and the p 's. The solution is unique up to adding any solutions of the associated homogeneous equation

$$V_m[\phi_j, \phi_m] = 0 \quad (\text{B.14})$$

We denote all the linearly independent solutions to (B.14) by $V_{am}(q, p)$, $a = 1, \dots, A$. The most general solution to (B.13) can be written as

$$u_m = U_m + v_a V_{am} \quad (\text{B.15})$$

We introduce the total Hamiltonian

$$H_T = H' + v_a \phi_a \quad H' = H + U_m \phi_m, \quad \phi_a = V_{am} \phi_m \quad (\text{B.16})$$

The equation of motion of any dynamical variable read simply

$$\dot{F} \approx [F, H_T] \quad (\text{B.17})$$

As a result of this analysis, we have satisfied all the consistency requirements of the theory and we still have arbitrary coefficients v , which implies that the general solution of the equation of motion contains arbitrary functions even if the initial conditions are completely fixed. The number of the arbitrary functions is equal to the number of primary constraints. This provides a difference of the generalized Hamiltonian formalism from what one is familiar with in elementary dynamics. These arbitrary functions in time implies that we are using a mathematical framework containing arbitrary features, for example, a coordinate system which we can choose in some arbitrary way, or the gauge in the electrodynamics.

Let us introduce some terminology which will enable one to appreciate the relationships between the quantities occur in the formalism. Any dynamical variable, R , a function of q 's and p 's is called *first-class* if it has zero Poisson brackets with all the ϕ 's

$$[R, \phi_j] \approx 0 \quad (\text{B.18})$$

Otherwise, R is *second-class*.

An important feature of the first-class property is that the Poisson bracket of two first-class functions is still first-class. We prove it below. Suppose F and G are first-class, we have

$$[F, \phi_j] = f_{j,j'} \phi_{j'}, \quad [G, \phi_j] = g_{j,j'} \phi_{j'} \quad (\text{B.19})$$

Now, by the Jacobi identity, we have

$$[[F, G], \phi_j] = [F, [G, \phi_j]] - [G, [F, \phi_j]] \approx 0 \quad (\text{B.20})$$

Finally, let us try to get a physical understanding of the situation where we start with given a initial state and get a solution containing arbitrary functions. The initial variables are q 's and p 's. We do not need initial values for the coefficients v . These initial conditions describe a physical state.

Now, the initial state must determine the state at later times. However, the q 's and p 's at later times are not uniquely determined by the initial state because we have the arbitrary functions v coming in. Instead, there must be several choices of q 's and p 's which correspond to the same state. So we have the problem of looking for all the sets of q 's and p 's that correspond to one particular physical state. For a general dynamical variable g , if we choose two different coefficients v and v' , the value of g at time δt differs by

$$\Delta g(\delta t) = \delta t(v_a - v'_a)[g, \phi_a] = \epsilon_a[g, \phi_a] \quad (\text{B.21})$$

where $\epsilon_a = \delta t(v_a - v'_a)$ is a small arbitrary number, small because of δt and arbitrary because of v . Thus, we come to the conclusion that the ϕ_a 's appearing in the theory generates infinitesimal transformation that does not affect the physical state.

These are not the only generators of the gauge transformations. We can go on further and find another two types:

- Apply two of these infinitesimal transformations $\epsilon_a \phi_a$ and $\gamma_a \phi_a$ in succession. Applying in different orders will give a difference (after using the Jacobi identity)

$$\epsilon_a \gamma_{a'}[g, [\phi_a, \phi_{a'}]] \quad (\text{B.22})$$

Thus, we can also use $[\phi_a, \phi_{a'}]$ as a generating function of an infinitesimal transformation that does not change the physical state. The only way these transformations (obtained from the Poisson bracket) are more general than ones we had before is that they can be secondary constraints.

- The Poisson bracket $[\phi_a, H']$ of any first-class primary constraint ϕ_a and the first-class Hamiltonian H' also generate a gauge transformation. We prove it as follows. We compare the values of the dynamical variable g at time $t + \delta t$ obtained by (i) first making a gauge transformation of parameter v_a and then evolving the system with H' and (ii) doing the same operations in the reverse order. The net difference is (after using the Jacobi identity)

$$\Delta g = \delta t v_a [g, [\phi_a, H']] \quad (\text{B.23})$$

which must be a gauge transformation.

However, it is not possible to infer from these considerations that every first-class secondary constraint generates a gauge transformation. One can actually construct such examples. Nevertheless, one postulates, in general, that all first-class constraints generate gauge transformations.

B.2 Example: Maxwell theory

As an illustration to the abstract theory, let us consider the Maxwell theory

$$S = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} d^4x \quad (\text{B.24})$$

The canonical momentum of the canonical coordinate A_μ is

$$B^\mu = \frac{\delta S}{\delta A_\mu} = F^{\mu 0} \quad (\text{B.25})$$

By taking $\mu = 0$, we get the primary constraint

$$B^0 \approx 0 \quad (\text{B.26})$$

These momenta will satisfy the (equal-time) Poisson bracket

$$[A_\mu(x), B^\nu(y)] = g_\mu^\nu \delta^{(3)}(x - y) \quad (\text{B.27})$$

where x, y are spatial coordinates.

Let us now introduce the Hamiltonian. We define that in the usual way by

$$\begin{aligned} H &= \int B^\mu \partial_0 A_\mu d^3x - L \\ &= \int \left(B^0 A_0 + F^{r0} \partial_0 A_r + \frac{1}{4} F_{rs} F^{rs} + \frac{1}{2} F_{r0} F^{r0} \right) d^3x \\ &= \int \left(B^0 A_0 + \frac{1}{4} F_{rs} F^{rs} - \frac{1}{2} F^{r0} F_{r0} + F^{r0} \partial_r A_0 \right) d^3x \\ &= \int \left(\frac{1}{4} F_{rs} F^{rs} + \frac{1}{2} B_r B_r - A_0 (\partial_r B_r - B^0) \right) d^3x \end{aligned} \quad (\text{B.28})$$

We can also utilize the ambiguity of H to get rid of the term $A_0 B^0$ and have

$$H = \int \left(\frac{1}{4} F_{rs} F^{rs} + \frac{1}{2} B_r B_r - A_0 \partial_r B_r \right) d^3x \quad (\text{B.29})$$

Since $B^0 \approx 0$ has to remain satisfied at all times, $[B^0, H]$ has to be zero, which leads to the secondary constraint

$$\partial_r B_r \approx 0 \quad (\text{B.30})$$

One can check that $[\partial_r B_r, H] = 0$ reduces to $0 = 0$ and thus does not give us anything new.

We now have to check whether they are first or second-class. In this example, they are all first-class. Our expression for the Hamiltonian is also first-class. The total Hamiltonian is

$$H_T = \int \left(\frac{1}{4} F_{rs} F^{rs} + \frac{1}{2} B_r B_r \right) d^3x - \int A_0 \partial_r B_r d^3x + \int v(x) B_0 d^3x \quad (\text{B.31})$$

The equation of motion of A_0 is

$$\partial_0 A_0 = v \quad (\text{B.32})$$

This gives us a meaning for the arbitrary coefficient v .

To get the most general motion, which is physically permissible, we add the secondary constraint with arbitrary coefficients

$$H_E = H_T + \int u(x) \partial_r B_r d^3x \quad (\text{B.33})$$

There is a certain simplification that is possible. This simplification comes from the fact that the variables A_0, B_0 are not of any physical significance. The equation of motion of B_0 is $B_0 = 0$ at all the times, which is not of interest. A_0 is something whose time derivative is quite arbitrary, which is again not of interest. Thus, we can drop them out from the theory and that will lead to a simplified Hamiltonian formalism with few degrees of freedom

$$H = \int \left(\frac{1}{4} F_{rs} F^{rs} + \frac{1}{2} B_r B_r \right) d^3x + \int u'(x) \partial_r B_r d^3x \quad (\text{B.34})$$

where we have set $v = 0$ and absorb A_0 into u .

One can impose gauge constraints to turn the first-class constraints into second class. For example, removing A_0, B_0 amounts to the gauge choice $A_0 = 0$. We can further impose the Coulomb gauge

$$\partial_r A_r = 0 \quad (\text{B.35})$$

so that $\partial_r B_r = 0$ together become second-class.

B.3 Quantization

To discuss the question of quantization, let us first take the case when there are no second-class constraints. We promote our coordinates and momenta into operators that satisfy the canonical commutation relations which correspond to the Poisson bracket relations of the classical theory. Then we impose the constraint on the Hilbert space, namely

$$\phi_j \psi = 0 \quad (\text{B.36})$$

The consistency of the constraints requires

$$[\phi_j, \phi_k] \psi = 0 \quad (\text{B.37})$$

We do not want to have any fresh conditions on ψ (this is ad-hoc). We want all the conditions on ψ to be included among (B.36), which means that we require

$$[\phi_j, \phi_k] = c_{jk,\ell} \phi_\ell \quad (\text{B.38})$$

Since ϕ_j 's are all first-class in the classical theory, their Poisson bracket is a linear combination of ϕ 's. However, for the quantum commutator, it does not necessarily follow that the coefficients c are all on the left because q 's and p 's do not commute. When we set up the constraints ϕ 's in the quantum theory, there may be some arbitrariness of how to order q 's and p 's. We have to try to arrange the order such that (B.38) holds.

There is a similar discussion needed in order to check that the constraints are consistent with the Schrodinger equation. If we start with a state that satisfies the constraint, it must remain to do so during the entire time evolution. This requires

$$[\phi_j, H] = b_{jk} \phi_k \quad (\text{B.39})$$

Let us now consider the case where there are only second-class constraints. The simplest example of two second-class constraints is

$$q_1 \approx 0, \quad p_q \approx 0 \quad (\text{B.40})$$

If we naively promote the Poisson bracket to the quantum commutator, we immediately run into contradiction

$$[q_1, p_1]\psi = i\psi = 0\psi \quad (\text{B.41})$$

In this simple case, it is obvious what the solution is. The variables q_1 and p_1 are not of interest if they are both set to zero. We have to throw them away to do quantization. The existence of second-class constraints means that there are some degrees of freedom which are not physically important. We have to pick out these degrees of freedom and set up new Poisson brackets referring only to the other degrees of freedom which are of physical importance. In terms of those new Poisson brackets, we can pass over to the quantum theory.

In the presence of both first-class and second-class constraints, we have to first do linear combination to have as many constraints as possible brought into the first class. And treat the residual second-class by introducing the Dirac bracket. It takes care of the constraints properly and does not change the equation of motion, i.e.

$$\begin{aligned} [g, H_T]_D &= [g, H_T] - [g, \chi_s]c_{ss'}[\chi_{s'}, H_T] \\ &\approx [g, H_T] \end{aligned} \quad (\text{B.42})$$

which follows from the fact that H_T is first-class.

In the quantum theory, the distinction between primary and secondary ceases to be important. The distinction between first-class and second-class is very important. We must put as many as possible into the first class and bring in the Dirac bracket which enables us to treat the surviving second-class constraints properly.

C Edge theory of the Abelian Chern-Simons theory

The Abelian Chern-Simons theory, in its full generality, can be written as

$$S = \frac{1}{4\pi} \int_Y K_{IJ} A^I dA^J \quad (\text{C.1})$$

where K_{IJ} is a symmetric invertible matrix with *integer* entries and A^I 's are $U(1)$ gauge fields. This action is useful in the study of the hierarchy states of fractional quantum hall systems, and is called the K-matrix formalism. For purely bosonic theories, the diagonal elements of K must be all even integers. For fermionic theories, at least one diagonal element is an odd integer.

Now, we consider the theory on a spatial disk, i.e., the space-time manifold is $Y = D^2 \times R$. By following the recipe discussed in Sec.3.1, we can write down the effective action for the boundary modes

$$S = \frac{1}{4\pi} \int_{\partial Y} K_{IJ} \partial_\sigma \phi^I \partial_t \phi^J - V_{IJ} \partial_\sigma \phi^I \partial_\sigma \phi^J \quad (\text{C.2})$$

where V_{IJ} is a positive definite constant matrix that depends on the choice of the boundary condition and is non-universal. The boundary field ϕ is related to the boundary value of the field A_σ by

$$A_\sigma \Big|_{\partial Y} = \partial_\sigma \phi \quad (\text{C.3})$$

We will first explain the canonical quantization of Eq. (C.2) and use the $U(1)_k$ Abelian Chern-Simons theory as the simplest interesting example to discuss various aspects. At the end, we will extend the discussion to the general case.

C.1 Canonical quantization

As we have discussed in Sec. 3.2, there are two equivalent approach to find the canonical formalism and canonical quantization. In the context of the non-Abelian Chern-Simons theory, we only use one approach due to the non-linearity of the configuration space. Here, we will do it with both approaches and explicitly show their equivalence.

Dirac's approach We treat ϕ^I as the canonical coordinate of the phase space and define the conjugate momentum by

$$\Pi_J(\sigma) := \frac{\delta L}{\delta \dot{\phi}^J} = \frac{1}{4\pi} K_{IJ} \partial_\sigma \phi^I \quad (\text{C.4})$$

The Poisson bracket is defined as usual

$$\{\phi^I(\sigma), \Pi_J(\sigma')\}_P := \delta_{IJ} \delta(\sigma - \sigma') \quad (\text{C.5})$$

The definition of the conjugate momentum yields primary constraints

$$\chi_I(\sigma) = \Pi_I(\sigma) - \frac{1}{4\pi} K_{IJ} \partial_\sigma \phi^J = 0 \quad (\text{C.6})$$

It turns out that these are the only constraints and they fall into the second class. Their Poisson brackets are given by

$$\begin{aligned} C_{IJ}(x, y) &= \{\chi_I(x), \chi_J(y)\}_P = \{\Pi_I, -\frac{K_{JM}}{4\pi} \partial_y \phi^M(y)\}_P + \{-\frac{K_{IM}}{4\pi} \partial_x \phi^M(x), \Pi_J(y)\}_P \\ &= -\frac{1}{2\pi} K_{IJ} \partial_x \delta(x - y) \end{aligned} \quad (\text{C.7})$$

The inverse of the above C function is

$$(C^{-1})_{IJ}(x, y) = -2\pi(K^{-1})_{IJ}(\theta(x - y) + \text{const}) \quad (\text{C.8})$$

where $\theta(x)$ is the step function that jumps between 0 and 1, const is an undetermined constant. One can check that the above C^{-1} indeed satisfies

$$\begin{aligned} \int C_{IJ}(x, y)(C^{-1})_{JK}(y, z) dy &= \delta_{IK} \int \partial_x \delta(x - y) (\theta(y - z) + \text{const}) dy \\ &= \delta_{IK} \partial_x (\theta(x - z) + \text{const}) \\ &= \delta_{IK} \delta(x - z) \end{aligned}$$

Then, we can define the Dirac bracket between any two functions on the phase space as

$$\{f, g\}_D := \{f, g\}_P - \{f, \chi_I\}_P C_{IJ}^{-1} \{\chi_J, g\}_P \quad (\text{C.9})$$

In particular, we are interested in the Dirac bracket of ϕ^I

$$\{\phi^I(x), \phi^J(y)\}_D = -2\pi(K^{-1})_{IJ} \theta(x - y) \quad (\text{C.10})$$

Witten's approach The action itself is already in the Hamiltonian form (first order in time derivative), and it is more convenient to shortcut Dirac's procedure. Let us consider the variation of the action

$$\delta S = - \int d\sigma dt \frac{1}{2\pi} K_{IJ} \delta\phi^I \partial_\sigma \partial_t \phi^J \quad (\text{C.11})$$

which yields an anti-symmetric matrix

$$F_{IJ}(x, y) = -\frac{1}{2\pi} K_{IJ} \delta(x - y) \partial_y \quad (\text{C.12})$$

If we regard ϕ^I as the phase space variables, we can directly use the inverse of F_{IJ} to define the Poisson bracket as

$$\{\phi^I(x), \phi^J(y)\}_P := (F^{-1})_{IJ}(x, y) = -2\pi(K^{-1})_{IJ}(\theta(x - y) + \text{const}) \quad (\text{C.13})$$

We get the same result.

Quantization The canonical quantization is simply to replace the bracket by the quantum commutator, which in this case is

$$[\phi^I(x), \phi^J(y)] = -2\pi i (K^{-1})_{IJ}(\theta(x - y) + \text{const}) \quad (\text{C.14})$$

Note that this commutator is non-local. This is not problematic because ϕ is a compact variable and is not a well-defined local operator by itself. Its derivative is a well-defined local operator and has the commutator

$$[\partial_x \phi^I(x), \partial_y \phi^J(y)] = 2\pi i (K^{-1})_{IJ} \delta'(x - y) \quad (\text{C.15})$$

C.2 Example: $U(1)_k$

Let us begin with the simplest interesting example, the $U(1)_k$ Chern-Simons theory, and discuss with details. This theory can describe the Laughlin states. There is only one $U(1)$ gauge field A . The bulk and the edge (with Lorentzian invariant boundary condition) action is

$$\begin{aligned} S[A] &= \frac{k}{4\pi} \int_Y A \wedge dA \\ S[\phi] &= \frac{k}{4\pi} \int_{\partial Y} \partial_\sigma \phi \partial_t \phi - (\partial_\sigma \phi)^2, \quad \phi \simeq \phi + 2\pi \end{aligned} \quad (\text{C.16})$$

The gauge field and the boundary field are still related by $A_\sigma = \partial_\sigma \phi$. The boundary condition of ϕ is related to the holonomy of the gauge field and thus is

$$\int_0^{2\pi} \partial_\sigma \phi d\sigma = 2\pi Q, \quad Q \in \mathbb{Z} \quad (\text{C.17})$$

Here, we require Q to be an integer in order to be consistent with the absence of topological defect in the bulk (i.e., the A has a trivial holonomy $\langle e^{i \oint A_\sigma} \rangle = 1$). In the context of fractional

quantum Hall systems, $\frac{1}{2\pi}\partial_\sigma\phi$ is the electron density and Q is the number of electrons on the edge. The canonical quantization yields the following commutation relation

$$\begin{aligned} [\phi(x), \phi(y)] &= -\frac{2\pi i}{k}\theta(x-y) \\ [\partial_x\phi(x), \partial_y\phi(y)] &= \frac{2\pi i}{k}\delta'(x-y) \end{aligned} \quad (\text{C.18})$$

where we have chosen the undetermined constant to be zero [any reason for this choice, or we should treat it as part of the definition?].

Local operators First, let us determine the operator content in the theory. In addition to the derivatives of ϕ , local operators at least have to be the vertex operators

$$: e^{in\phi(x)} :, \quad n \in \mathbb{Z} \quad (\text{C.19})$$

In order to determine whether they are actually local or not, we can compute their commutator with $\partial_\sigma\phi$ and among themselves. These commutators can be computed using the Baker-Campbell-Hausdorff formula (the normal ordering does not matter in this calculation):

- The commutation with $\partial_x\phi$

$$[\partial_x\phi(x), : e^{in\phi(y)} :] = \frac{2\pi n}{k}\delta(x-y) : e^{in\phi(y)} : \quad (\text{C.20})$$

implies that $: e^{in\phi(y)} :$ can modify the boundary condition of ϕ , i.e.

$$\left[\int_0^{2\pi} \partial_x\phi(x) dx, : e^{in\phi(y)} : \right] = \frac{2\pi n}{k} : e^{in\phi(y)} : \quad (\text{C.21})$$

The twist is restricted to be an integral multiple of 2π , and thus only

$$: e^{iNk\phi} :, N \in \mathbb{Z}$$

act within the Hilbert space.

- The commutators among the vertex operator themselves are

$$\begin{aligned} : e^{in\phi(x)} :: e^{in'\phi(y)} : &= e^{-nn'[\phi(x), \phi(y)]} : e^{in'\phi(y)} :: e^{in\phi(x)} : \\ &= e^{\frac{2\pi inn'}{k}\theta(x-y)} : e^{in'\phi(y)} :: e^{in\phi(x)} : \end{aligned} \quad (\text{C.22})$$

Not all the vertex operators are mutually local, only

$$: e^{iNk\phi} :, N \in \mathbb{Z}$$

commute with other ones and can be treated as local operators. Furthermore, We notice that $: e^{ik\phi}$ is actually a fermionic operator when k is odd.

Therefore, the set of local operators are generated by

$$\partial_x^n \phi, \quad : e^{\pm ik\phi} : \quad (\text{C.23})$$

In the context of the fractional quantum Hall systems, they describes the density fluctuation and adding/removing an electron on the edge.

Spectrum Now, we determine the energy spectrum of the theory. We will use the spectrum to determine the partition function and identify it with the chiral character of the compact boson.

The quantum Hamiltonian is defined with the normal ordering in order to remove the diverging ground state energy

$$H = \frac{k}{4\pi} \int_0^{2\pi} : (\partial_\sigma \phi)^2 : d\sigma \quad (\text{C.24})$$

We can separate ϕ into two parts, the zero mode which includes the winding and the quantum part which includes the fluctuation

$$\phi(\sigma) = \phi_0 + Qx + \phi_{qu}(\sigma) \quad (\text{C.25})$$

where $\phi_{qu}(\sigma)$ is periodic in space and has the following Fourier expansion

$$\phi_{qu}(\sigma) = \frac{1}{\sqrt{2\pi}} \sum_{n \neq 0} \phi_n e^{-in\sigma} \quad (\text{C.26})$$

The commutation relation of ϕ_n is

$$[\phi_n, \phi_m] = \frac{2\pi}{k} \frac{1}{n} \delta_{n+m} \quad (\text{C.27})$$

We want to treat $\phi_{n>0}$ as the annihilation operator and the Hamiltonian in terms of the Fourier mode is

$$H = \frac{k}{2} Q^2 + \frac{k}{2\pi} \sum_{n>0} n^2 \phi_{-n} \phi_n \quad (\text{C.28})$$

The Hilbert space falls into topological sectors that are labeled by Q . The vertex operator $e^{\pm ik\phi}$ maps state from one sector to the other. Each sector has a ground state that satisfies

$$\phi_{n>0} |Q; 0\rangle = 0 \quad (\text{C.29})$$

The excited states are created by applying ϕ_{-n} (or equivalently $\partial^k \phi$).

The partition function is obtained by summing over states from different topological sectors. The result is given by

$$Z(\tau) = \frac{1}{\eta(\tau)} \sum_{Q \in \mathbb{Z}} q^{\frac{k}{2} Q^2} \quad (\text{C.30})$$

where $\eta(\tau)$ comes from summing over the states within each topological sector.¹⁵ This is the character of the compact boson with radius $R^2 = k$ (and $\alpha' = 2$ in Polchinski's convention). More generally, let us define the chiral character for the twist sectors as

$$\chi_\ell(\tau) = \frac{1}{\eta(\tau)} \sum_{Q \in \mathbb{Z}} q^{\frac{1}{2k}(kQ+\ell)^2}, \quad 0 \leq \ell < k \quad (\text{C.31})$$

¹⁵Strictly speaking, the normal-ordered Hamiltonian can not give us the $q^{1/24}$ term in $\eta(\tau)$.

Its modular transformation is

$$\chi_\ell(\tau + 1) = e^{2\pi i \left(\frac{\ell^2}{2k} - \frac{c}{24}\right)} \chi_\ell(\tau), \quad \chi_\ell(-1/\tau) = \frac{1}{\sqrt{k}} \sum_{\ell'} e^{\frac{2\pi i \ell \ell'}{k}} \chi_{\ell'}(\tau) \quad (\text{C.32})$$

Thus, the partition function is not modular invariant. The detail of this computation will be provided later when we discuss the partition function with defect lines.

Correlators Finally, let us look at some Euclidean correlation functions. This can be easily computed by using the Heisenberg operator (we suppress the zero mode, which is important in the computation of the correlation function of vertex operators)

$$\phi(x, \tau) = e^{H\tau} \phi(x) e^{-H\tau} = \frac{1}{\sqrt{2\pi}} \sum_{n \neq 0} \frac{\phi_n}{z^n} \quad z = e^{\tau + ix} = e^w \quad (\text{C.33})$$

Then one can show that

$$\langle \partial_x \phi(x, \tau) \partial_x \phi(0, 0) \rangle = \frac{1}{k} \frac{1}{(2 \sinh \frac{w}{2})^2} \quad (\text{C.34})$$

When the spatial dimension becomes large, the result gets simplified

$$\langle \partial_x \phi(x, \tau) \partial_x \phi(0, 0) \rangle_{plane} = \frac{1}{k} \frac{1}{(\tau + ix)^2} \quad (\text{C.35})$$

Accordingly, we have

$$\langle \phi(x, \tau) \phi(0, 0) \rangle_{plane} = -\frac{1}{k} \log(ix + \tau) \quad (\text{C.36})$$

$$\langle e^{in\phi(x, \tau)} e^{-in\phi(0, 0)} \rangle_{plane} = \frac{1}{(ix + \tau)^{n^2/k}} \quad (\text{C.37})$$

The vertex operator $e^{in\phi}$ thus has the conformal dimension

$$h_n = \frac{n^2}{2k} \quad (\text{C.38})$$

This is equivalent to Polchinski's convention at $\alpha' = 2, R^2 = k$. When $n = k$, the conformal dimension is $k/2$, which is a half integer for odd k . This is consistent with the fact that $e^{ik\phi}$ is a fermionic operator at odd k . When $n = kQ$ with $Q \in \mathbb{Z}$, we see that the conformal dimension of $e^{ikQ\phi}$ is the same as the ground state energy of the charge Q sector. This is consistent with the state-operator correspondence.

Partition function with defect line The global U(1) symmetry allows us to introduce partition function with a defect line, which in this case is a chemical potential

$$Z(\tau, \mu) := \text{Tr}_{\mathcal{H}} q^{\widehat{L}_0 - \frac{1}{24}} e^{i\mu \widehat{Q}}, \quad q = e^{2\pi i \tau}, \quad \mu \in [0, 2\pi] \quad (\text{C.39})$$

where we use $\widehat{}$ to emphasis that $\widehat{L}_0$ and especially $\widehat{Q}$ are operators. Such an object is quite useful in the study of modular bootstrap. More generally, we can define the partition function with a defect line in the twisted sector

$$Z_\ell(\tau, \mu) := \text{Tr}_{\mathcal{H}_\ell} q^{\widehat{L}_0 - \frac{1}{24}} e^{i\mu \widehat{Q}} \quad (\text{C.40})$$

Our goal here is to compute the modular S -matrix of this object and extract its high temperature limit ($\tau \rightarrow 0$), which is related to the charged Cardy formula.

The expression of Z_ℓ follows from the previous results about the spectrum and is a simple modification to (C.31)

$$Z_\ell(\tau, \mu) = \frac{1}{\eta(\tau)} \sum_{Q \in \mathbb{Z}} q^{\frac{1}{2k}(\ell + kQ)^2} e^{i\mu(Q + \frac{\ell}{k})}$$

Note that the extra $i\mu\ell/k$ term is added because the boundary condition of ϕ is shifted by ℓ/k in the twisted sector. This term is related to the fractional charge of the anyon in the bulk. Its modular S -transformation can be computed by the Poisson summation formula. The computation is detailed below

$$\begin{aligned} Z_\ell(-\frac{1}{\tau}, \mu) &= \frac{e^{i\mu\ell/k}}{\sqrt{-i\tau}\eta(\tau)} \sum_{Q \in \mathbb{Z}} \exp \left[-\frac{2\pi i}{\tau} \frac{(\ell + kQ)^2}{2k} + i\mu Q \right] \\ &= \frac{e^{i\mu\ell/k}}{\sqrt{-i\tau}\eta(\tau)} \int_{\mathbb{R}} dx \sum_{M \in \mathbb{Z}} \exp \left[2\pi i M x - \frac{2\pi i}{\tau} \frac{(\ell + kx)^2}{2k} + i\mu x \right] \\ &= \frac{e^{i\mu\ell/k}}{\sqrt{-i\tau}\eta(\tau)} \int_{\mathbb{R}} dx \sum_{M \in \mathbb{Z}} \exp \left[2\pi i \left(\left(M + \frac{\mu}{2\pi} \right) x - \frac{k}{2\tau} \left(x + \frac{\ell}{k} \right)^2 \right) \right] \end{aligned}$$

We can complete the square as

$$\begin{aligned} & \left(M + \frac{\mu}{2\pi} \right) x - \frac{k}{2\tau} \left(x + \frac{\ell}{k} \right)^2 \\ &= -\frac{k}{2\tau} \left(x + \frac{\ell}{k} - \left(M + \frac{\mu}{2\pi} \right) \frac{\tau}{k} \right)^2 - \frac{\ell}{k} \left(M + \frac{\mu}{2\pi} \right) + \frac{\tau}{2k} \left(M + \frac{\mu}{2\pi} \right)^2 \end{aligned}$$

Then, we can perform the Gaussian integral in x first and have

$$Z_\ell(-\frac{1}{\tau}, \mu) = \frac{1}{\sqrt{k}} \frac{1}{\eta(\tau)} \sum_{M \in \mathbb{Z}} e^{-2\pi i \frac{\ell M}{k}} q^{\frac{1}{2k} \left(M + \frac{\mu}{2\pi} \right)^2} \quad (\text{C.41})$$

Let $M = \ell' + Q'k$ with $\ell' = 0, 1, \dots, k-1$ and $Q' \in \mathbb{Z}$, we can reorganize the summation as a double sum

$$Z_\ell(-\frac{1}{\tau}, \mu) = \frac{1}{\sqrt{k}} \frac{1}{\eta(\tau)} \sum_{\ell'=0}^{k-1} \sum_{Q' \in \mathbb{Z}} e^{-\frac{2\pi i}{k} \ell \ell'} q^{\frac{1}{2k} (\ell' + Q'k + \frac{\mu}{2\pi})^2}$$

We can let $\ell' = k - \tilde{\ell}$ and modifies Q' accordingly to get the following final form

$$Z_\ell(-\frac{1}{\tau}, \mu) = \frac{1}{\sqrt{k}} \frac{1}{\eta(\tau)} \sum_{\tilde{\ell}=1}^k \sum_{\tilde{Q} \in \mathbb{Z}} e^{\frac{2\pi i}{k} \tilde{\ell} \tilde{\ell}} q^{\frac{1}{2k} (\tilde{\ell} + \tilde{Q}k - \frac{\mu}{2\pi})^2} \quad (\text{C.42})$$

Written more compactly, we have

$$Z_\ell(-\frac{1}{\tau}, \mu) = \exp \left(\frac{i\tau\mu^2}{4\pi k} \right) \sum_{\tilde{\ell}=1}^k S_{\ell, \tilde{\ell}} Z_{\tilde{\ell}}(\tau, -\tau\mu), \quad S_{\ell, \tilde{\ell}} = \frac{1}{\sqrt{k}} e^{\frac{2\pi i}{k} \ell \tilde{\ell}} \quad (\text{C.43})$$

Note that the chemical potential is also changed under the modular transformation.

Now, we can extract the high-temperature limit ($\tau \rightarrow 0$) of the partition function $Z(\tau, \mu)$. If we assume $-\pi < \mu < \pi$, the leading term is given by

$$Z(\tau, \mu) \sim \frac{1}{\sqrt{k}} \exp\left(-\frac{i\mu^2}{4\pi k\tau}\right) \tilde{q}^{-\frac{1}{24}} = \frac{1}{\sqrt{k}} \exp\left[\left(\frac{\pi}{12} - \frac{\mu^2}{4\pi k}\right) \frac{L}{\epsilon}\right] \quad (\text{C.44})$$

In the second step, we choose a particular modular parameter $\tau = i\epsilon/L$. This is related to the charged Cardy formula. When $\mu = \pi$, both $\ell = 0$ and $\ell = 1$ make contribution, the leading term is amplified by a factor of 2.

The above result is mixed with the background thermal fluctuation. We can isolate the contribution of the defect line by considering the ratio

$$\ln \langle e^{i\mu\hat{Q}} \rangle := \ln \frac{Z(\tau, \mu)}{Z(\tau, 0)} \sim -\frac{\mu^2}{4\pi k} \frac{L}{\epsilon} - \frac{1 + \delta_{\mu, \pi}}{2} \ln k \quad (\text{C.45})$$

Let us give a physical interpretation of this result using a different context.¹⁶ Suppose that we have a two-dimensional $U(1)$ symmetric gapped state and consider the expectation value of $e^{i\mu\hat{Q}_D}$, where $\hat{Q}_D$ is the total charge associated to a disk subregion D . Since the state preserves the $U(1)$ symmetry, $e^{i\mu\hat{Q}_D}$ acts non-trivially only near the edge of the subregion and can be approximated by a line operator that is supported on ∂D . The line operator should be regarded as a defect line because any charged object with a charge q will get a phase factor $e^{i\mu q}$ after encircling the line. Generically, the line operator has a line tension and its expectation value satisfies a perimeter law

$$\ln \langle e^{i\mu Q_D} \rangle = -\alpha L_D - \gamma \quad (\text{C.46})$$

Then, (C.45) implies that if the low-energy physics of the gapped state is described by the $U(1)_k$ Chern-Simons theory, we have

$$\alpha = \frac{\mu^2}{4\pi} \frac{1}{k} \frac{1}{\epsilon}, \quad \gamma = \frac{1 + \delta_{\mu, \pi}}{2} \ln k \quad (\text{C.47})$$

The constant correction is *universal* and related to the level of the Chern-Simons theory. The line tension has more structures: it is quadratic in μ , has a UV divergence from $1/\epsilon$ and proportional to the level. However, this case is too special that it obscures the general structure of the line tension. In the following, we will discuss the Abelian Chern-Simons theory with a general K -matrix and come back to this point, where the correct interpretation becomes more clear.

C.3 General K -matrix

In this section, we return to the general case. It is technically not that different from the simplest $U(1)_k$ Chern-Simons theory and thus we will mostly quote results without proof. The action is copied below

$$S = \frac{1}{4\pi} \int_Y K_{IJ} a^I da^J, \quad I = 1, \dots, N$$

¹⁶We thank Meng Cheng and Yingfei Gu for helpful private discussion.

Here, we use the lower-case a^I for the dynamical $U(1)$ gauge fields and save the upper case A for the external $U(1)$ gauge field that is introduced below.

We want to specify an *external* global $U(1)$ symmetry of the system, that is coupled to the *external* electromagnetic fields, to eventually make some contact to the fractional quantum hall systems. However, such a choice is not canonical and can be denoted by a charge vector t with integer components. Thus, the action with the $U(1)$ background electromagnetic field is

$$S[a, A] = \frac{1}{4\pi} \int_Y K_{IJ} a^I da^J - \frac{1}{2\pi} \int t_I A da^I \quad (\text{C.48})$$

Bulk properties We first list some bulk properties of the theory:

- Hall conductance

$$\sigma_H = \frac{1}{2\pi} t^T K^{-1} t \quad (\text{C.49})$$

- The anyons are described by a bosonic current j and characterized by another integer vector ℓ as follows

$$\int \ell_I a^I \cdot j \quad (\text{C.50})$$

Such an anyon carries a $U(1)$ charge

$$q_\ell = \ell^T K^{-1} t \quad (\text{C.51})$$

- Self-statistics

$$\theta_{\ell, \ell} = \pi \ell^T K^{-1} \ell \quad (\text{C.52})$$

- Mutual braiding statistics

$$\theta_{\ell, \ell'} = 2\pi \ell^T K^{-1} \ell' \quad (\text{C.53})$$

- Local excitation ℓ ,

$$\ell = K \Lambda \quad (\text{C.54})$$

where Λ is an arbitrary integer vector. This condition guarantees that it has a trivial braiding statistics, i.e., $\theta_{\ell, \ell'} \in 2\pi\mathbb{Z}$ for all integer vectors ℓ' . A local excitation always carry an integer amount of charge $\Lambda^T t$.

- Two anyon vectors are equivalent if they differ by a local excitation

$$\ell \sim \ell' \quad \text{if} \quad \ell - \ell' = K \Lambda \quad \text{for some } \Lambda \quad (\text{C.55})$$

We will use $\mathcal{L}$ to denote the set of representative vectors from the equivalence classes. There is one pictorial way of representing the inequivalent (abelian) anyons. We think of the integer vectors ℓ that label the anyons as a vector in the N -dimensional integer lattice $\mathbb{Z}^N$. The primitive vectors are given by the columns of the K matrix,

$$\{\mathbf{b}_i \in \mathbb{Z}^N | i = 1, \dots, N, \mathbf{b}_{i,j} = K_{j,i}\}$$

which form a Bravais lattice. The distinct anyons thus correspond to the sublattices in this lattice. The fusion of anyons is the vector addition on this lattice. The statistics of two anyons represented by vectors $\mathbf{a}$ and $\mathbf{b}$ is determined by the inner product between the two vectors, where the inner product is defined as $\mathbf{a}^T K^{-1} \mathbf{b}$.

Example: We illustrate the above idea with one concrete example, the $\mathbb{Z}_2$ toric code, which is described by a 2×2 K matrix. There are four types of anyons $1, e, m, f$ and can be organized as follows on the anyon lattice

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \quad \begin{array}{c} \begin{array}{c} (0, 2) \\ \uparrow \\ m : (0, 1) \\ \bullet \\ 1 : (0, 0) \end{array} \quad \begin{array}{c} f : (1, 1) \\ \bullet \\ e : (1, 0) \end{array} \\ \begin{array}{c} \xrightarrow{(2, 0)} \end{array} \end{array} \quad (C.56)$$

If one were to make connection between the abstract theory and the actual quantum Hall system, it would have to satisfy two constraints: [8]

- Local excitations that carry an even $U(1)$ charge must be self bosons. It requires

$$t_I = K_{II} \mod 2 \quad (C.57)$$

- At least one local excitation is a fermion (electron). It requires

$$\gcd(t_1, \dots, t_N) = 1 \quad (C.58)$$

The following discussion will be general and not affected by whether imposing such constraints or not.

Edge properties The edge modes are also coupled to the background electromagnetic field. The action is

$$S[\phi, A] = \frac{1}{4\pi} \int K_{IJ} \partial_x \phi^I \partial_t \phi^J - V_{IJ} \partial_x \phi^I \partial_x \phi^J + \frac{1}{2\pi} \int \epsilon^{\mu\nu} t_I (\partial_\mu \phi^I) A_\nu \quad (C.59)$$

If we just turn on a chemical potential, $A_0 = \mu \delta(t)$ being a constant at a given time slice, the gauge coupling takes the form

$$\frac{\mu}{2\pi} \int t_I (\partial_x \phi^I) dx = \mu t_I Q^I \quad (C.60)$$

where $Q^I \in \mathbb{Z}$ is the boundary condition for the boundary field ϕ^I . Thus, we can identify $t_I Q^I$ as the total $U(1)$ charge of the system.

Generally, both chiral and anti-chiral modes coexist in the action. This can be made manifest by diagonalizing the K matrix. We introduce new variables $\phi = W \tilde{\phi}$, where W is a real $N \times N$ matrix that does

$$W^T K W = \Sigma = \begin{pmatrix} \mathbb{I}_n & \\ & -\mathbb{I}_{\tilde{n}} \end{pmatrix}, \quad n + \tilde{n} = N \quad (C.61)$$

The action written in terms of the new field variables $\tilde{\phi}$ is

$$\frac{1}{4\pi} \int \Sigma_{II} \partial_x \tilde{\phi}^I \partial_t \tilde{\phi}^I - \tilde{V}_{IJ} \partial_x \tilde{\phi}^I \partial_x \tilde{\phi}^J \quad (C.62)$$

We will assume that $\tilde{V}_{IJ} = \delta_{IJ}$ so that the dynamics of the system is described by a conformal field theory.¹⁷ It is then clear that there are n chiral modes and $\bar{n}$ anti-chiral modes, which corresponds to the eigenvectors of the K -matrix with positive and negative eigenvalues, respectively.

The local operators are the derivatives of ϕ^I and a subset of the vertex operators. A generic single-valued vertex operator is associated to an integer vector

$$: e^{i\ell^T \phi} :, \quad \ell_I \in \mathbb{Z} \quad (\text{C.63})$$

which corresponds to the anyon ℓ in the bulk. Let us compute its commutator with other operators using the baker-Campbell-Hausdorff formula:

- Commutator with $\partial_x \phi^I$

$$[\partial_x \phi^I(x), e^{i\ell^T \phi(y)}] = 2\pi(K^{-1})_{IJ} \ell_J \delta(x-y) e^{i\ell^T \phi(y)} \quad (\text{C.64})$$

implies that $e^{i\ell^T \phi(y)}$ changes the boundary condition of ϕ^I by $2\pi(K^{-1})_{IJ} \ell_J$ and thus carries a $U(1)$ charge $t^T K^{-1} \ell$. Since we require that the boundary conditions of all ϕ^I is an integral multiple of 2π (states carry integer charges), only vertex operators with $\ell = K\Lambda$ act within the Hilbert space.

- Commutator with another vertex operator

$$e^{i\ell_1^T \phi(x)} e^{i\ell_2^T \phi(y)} = e^{2\pi i \ell_1^T K^{-1} \ell_2 \theta(x-y)} e^{i\ell_2^T \phi(y)} e^{i\ell_1^T \phi(x)} \quad (\text{C.65})$$

Therefore, only vertex operators with $\ell = K\Lambda$ commute (or anti-commute) with other ones and can be regarded as local operators.

Therefore, the set of local operators are generated by the following ones

$$\partial_x^n \phi^I, \quad e^{i(K\Lambda)^T \phi}, \quad \Lambda_I \in \mathbb{Z} \quad (\text{C.66})$$

which, by the operator-state correspondence, generate all the states in the Hilbert space. In the quantum Hall systems, they describe the density fluctuation and creation/annihilation of fermions and bosons. Other vertex operators $e^{i\ell^T \phi}$, where ℓ is a representative of integer vectors that is not of the form of $K\Lambda$, give rise to the twisted sectors and correspond to creation/annihilation of anyons. Again, we use $\mathcal{L}$, the same as what we have introduced for the anyons, to denote the set of twisted sectors.

To accommodate the above discussion, let us write a generic vertex operator as

$$e^{i(\ell+K\Lambda)^T \phi}, \quad \ell \in \mathcal{L} \quad (\text{C.67})$$

In this free field theory, the conformal dimension of an operator is determined by the two-point function of the fundamental field, which can be read off from the equations of motion.

¹⁷The Hilbert space, as a representation of the local operator algebra, is not affected by the choice here. Because the algebra is determined by the quantization, which in this case is determined by K . The dynamics, which is related to the dispersion, is affected.

Therefore, to determine the conformal dimension of the vertex operator, it is more convenient to rewrite the vertex operator in terms of the variable $\tilde{\phi}$

$$e^{i\gamma^T \tilde{\phi}}, \quad \gamma = W^T(\ell + K\Lambda) \quad (\text{C.68})$$

Note that the first n components of $\tilde{\phi}$ are chiral and the last $\tilde{n}$ components are anti-chiral. The conformal dimension of the generic vertex operator is

$$h = \frac{1}{2}\gamma^T \begin{pmatrix} \mathbb{I}_n & \\ & 0 \end{pmatrix} \gamma, \quad \tilde{h} = \frac{1}{2}\gamma^T \begin{pmatrix} 0 & \\ & \mathbb{I}_{\tilde{n}} \end{pmatrix} \gamma \quad (\text{C.69})$$

The scaling dimension and the conformal spin are

$$\Delta = h + \tilde{h} = \frac{1}{2}\gamma^T \gamma, \quad s = h - \tilde{h} = \frac{1}{2}\gamma^T \Sigma \gamma = \frac{1}{2}\ell K^{-1} \ell + \text{integer} \quad (\text{C.70})$$

Note that the conformal spins of the vertex operators are the same as the self-statistics of the anyons, which is consistent with the bulk-edge correspondence.

Partition function with a defect line The above discussion prepares us for writing down the partition function in a twisted sector, labeled by ℓ . We also allow the insertion of a U(1) symmetry defect line. The general partition function is

$$Z_\ell[\tau, \mu] = \frac{1}{\eta(\tau)^n \bar{\eta}(\bar{\tau})^{\tilde{n}}} \sum_{\Lambda \in \mathbb{Z}^N} q^{h_{\ell+K\Lambda}} \bar{q}^{\tilde{h}_{\ell+K\Lambda}} e^{i\mu t^T K^{-1}(\ell+K\Lambda)}, \quad q = e^{2\pi i \tau} \quad (\text{C.71})$$

where we have used the fact that a vertex operator labeled by $\ell + K\Lambda$ carries the external U(1) charge $t^T K^{-1}(\ell + K\Lambda)$.

We are interested in the modular properties of the partition function. The modular T transformation is simple

$$Z_\ell(\tau + 1, \mu) = e^{2\pi i s_\ell} Z_\ell(\tau, \mu) \quad (\text{C.72})$$

where $s_\ell = \frac{1}{2}\ell^T K^{-1} \ell$ is the conformal spin of the vertex operator $e^{i\ell^T \phi}$. Since K is a non-degenerate matrix, only the identity sector $\ell = 0$ is modular T invariant. The modular S transformation is more complicated

$$Z_\ell(-\frac{1}{\tau}, \mu) = \exp\left(-\frac{\mu^2}{4\pi} t^T W A(\tau, \bar{\tau}) W^T t\right) \sum_{\ell' \in \mathcal{L}} S_{\ell, \ell'} Z_{\ell'}(\tau, \tilde{\mu}) \quad (\text{C.73})$$

where A is a diagonal matrix that is determined by the modular parameter

$$A(\tau, \bar{\tau}) = \begin{pmatrix} -i\tau \mathbb{I}_n & \\ & i\bar{\tau} \mathbb{I}_{\tilde{n}} \end{pmatrix} \quad (\text{C.74})$$

The chemical potential is also transformed and $\tilde{\mu}$ becomes a matrix

$$\tilde{\mu} = \mu W \begin{pmatrix} -\tau \mathbb{I}_n & \\ & -\bar{\tau} \mathbb{I}_{\tilde{n}} \end{pmatrix} W^{-1} \quad (\text{C.75})$$

The modular S matrix is related to the braiding statistics of the anyons in the bulk and is given by

$$S_{\ell, \ell'} = \frac{1}{\sqrt{|\det K|}} e^{i\theta_{\ell, \ell'}} \quad (\text{C.76})$$

One can check that it reduces the previous result in the $U(1)_k$ case.

The modular S transformation can be computed by the Poisson summation formula in high dimensions, which is reviewed in Appendix. D. Let us first introduce some notations. The states that differ by the $U(1)$ charge within the twist sector $\ell \in \mathcal{L}$ fit into the following Euclidean lattice

$$\Gamma_\ell = \{W^T(\ell + K\Lambda) | \Lambda \in \mathbb{Z}\} \quad (\text{C.77})$$

The basis vectors of the lattice correspond to the columns of the matrix $W^T K$ and thus the volume of the unit cell of the Euclidean lattice is

$$\text{vol}(\Gamma) = \sqrt{|\det K|} \quad (\text{C.78})$$

We can then rewrite the partition function as a summation over vectors in the lattice

$$Z_\ell[\tau, \mu] = \frac{1}{\eta(\tau)^n \bar{\eta}(\bar{\tau})^{\bar{n}}} \sum_{\gamma \in \Gamma_\ell} \exp \left[-\pi \gamma^T A(\tau, \bar{\tau}) \gamma + i \mu \tilde{t}^T \gamma \right] \quad (\text{C.79})$$

$$A(\tau, \bar{\tau}) = \begin{pmatrix} -i\tau \mathbb{I}_n & \\ & i\bar{\tau} \mathbb{I}_{\bar{n}} \end{pmatrix}, \quad \tilde{t} = \Sigma W^T t$$

Under a modular S -transformation, we have

$$Z_\ell\left(-\frac{1}{\tau}, \mu\right) = \frac{1}{\sqrt{|\det K|}} \frac{1}{\eta(\tau)^n \bar{\eta}(\bar{\tau})^{\bar{n}}} \sum_{p \in \Gamma_0^*} \exp \left[-\pi \left(p + \frac{\mu \tilde{t}}{2\pi}\right)^T A(\tau, \bar{\tau}) \left(p + \frac{\mu \tilde{t}}{2\pi}\right) - 2\pi i \ell^T W p \right] \quad (\text{C.80})$$

where Γ_0^* is the dual lattice of Γ_0 and we have used the property $A(\tau)^{-1} = A(-1/\tau)$. One can show that the dual lattice is equal to a union of the original lattices¹⁸

$$\Gamma_0^* = \bigcup_{\ell' \in \mathcal{L}} \Sigma \cdot \Gamma_{\ell'} \quad (\text{C.81})$$

¹⁸A vector $p \in \Gamma_0^*$ in the dual lattice should satisfy $p \cdot \gamma \in \mathbb{Z}, \forall \gamma \in \Gamma_0$. Since K and W are non-degenerate matrices, we can write $p = W^{-1} K^{-1} P = \Sigma W^T P$ without loss of generality. Then, for $\gamma = K^T K \Lambda \in \Gamma_0$, we have

$$p \cdot \gamma = P \cdot \Lambda \in \mathbb{Z}$$

which can be solved by imposing P to be an integer vector. The dual lattice Γ_0^* can be written as

$$\Gamma_0^* = \{\Sigma W^T \Lambda' | \Lambda' \in \mathbb{Z}^N\}$$

Note that any integer vector P can be written as in the form of $\ell' + K\Lambda'$ for some $\ell' \in \mathcal{L}$ and an integer vector Λ' . Therefore, we can rewrite the dual lattice as the union of the original lattices.

Therefore, the summation can be reorganized as a double sum

$$\sum_{p \in \Gamma_0^*} \stackrel{p=\Sigma\gamma}{=} \sum_{\ell' \in \mathcal{L}} \sum_{\gamma \in \Gamma_{\ell'}}$$

and we have (note that we changes the sign of ℓ' and γ to be consistent with the convention in the literature)

$$Z_{\ell}(-\frac{1}{\tau}, \mu) = \sum_{\ell' \in \mathcal{L}} \frac{1}{\sqrt{|\det K|}} \frac{1}{\eta(\tau)^n \bar{\eta}(\bar{\tau})^{\bar{n}}} \times \sum_{\gamma' \in \Gamma_{\ell'}} \exp \left[-\pi(\gamma - \frac{\mu}{2\pi} W^T t)^T A(\tau, \bar{\tau})(\gamma - \frac{\mu}{2\pi} W^T t) + 2\pi i \ell'^T K^{-1} \ell' \right] \quad (\text{C.82})$$

This completes the derivation of the modular S transformation (C.73) shown before.

Now, we consider the high-temperature limit ($\tau \rightarrow 0$) of the partition function. We choose a particle modular parameter $\tau = i\epsilon/L$ and let $\epsilon \rightarrow 0$, in which case the matrix A becomes proportional to the identity matrix. The leading term is

$$\langle e^{i\mu Q} \rangle := \ln \frac{Z(\tau, \mu)}{\ln Z(\tau, 0)} \sim -\frac{\mu^2}{4\pi} (t^T W W^T t) \frac{L}{\epsilon} - \frac{1}{2} \ln \det K \quad (\text{C.83})$$

One can show that the matrix $W W^T$ is almost the inverse of the K -matrix, with all the negative eigenvalues being flipped to become positive. Namely, if the K -matrix is diagonalized as

$$K = O \text{diag}(k_1, \dots, k_n, -k_{n+1}, \dots, -k_{n+\bar{n}}) O^T, \quad k_i > 0 \quad (\text{C.84})$$

Then we have

$$\tilde{K}^{-1} := W W^T = O \text{diag}(\frac{1}{k_1}, \dots, \frac{1}{k_n}, \frac{1}{k_{n+1}}, \dots, \frac{1}{k_{n+\bar{n}}}) O^T, \quad k_i > 0 \quad (\text{C.85})$$

and can write the result as

$$\langle e^{i\mu Q} \rangle := \ln \frac{Z(\tau, \mu)}{\ln Z(\tau, 0)} \sim -\frac{\mu^2}{4\pi} (t^T \tilde{K}^{-1} t) \frac{L}{\epsilon} - \frac{1}{2} \ln \det K \quad (\text{C.86})$$

This sign flip means that both the chiral and anti-chiral modes make positive contribution to the line tension

D Poisson summation formula

This section is a review of the Poisson summation formula in high dimensions. Let us consider a d -dimensional Euclidean lattice Γ

$$\Gamma = \{ \gamma_0 + \sum_{i=1}^d n_i \epsilon_i | n_i \in \mathbb{Z} \} \quad (\text{D.1})$$

where γ_0 is a vector lying inside the unit cell Γ_0

$$\Gamma_0 = \left\{ \sum_{i=1}^d x_i \epsilon_i \mid 0 \leq x_i < 1 \right\} \quad (\text{D.2})$$

The volume of the unit cell is given by

$$\text{vol}(\Gamma) = \sqrt{\det g_{ij}}, \quad g_{ij} = \epsilon_i \cdot \epsilon_j \quad (\text{D.3})$$

The dual lattice Γ^* is similarly defined

$$\Gamma^* = \left\{ \sum_{i=1}^d n_i \epsilon_i^* \mid n_i \in \mathbb{Z} \right\}, \quad \epsilon_i^* \cdot \epsilon_j = \delta_{ij} \quad (\text{D.4})$$

Note that there is no additional shift in the dual lattice. For an arbitrary scalar function $f(\gamma)$ defined on the lattice Γ , the Poisson summation formula is

$$\sum_{\gamma \in \Gamma} f(\gamma) = \frac{1}{\text{vol}(\Gamma)} \sum_{p \in \Gamma^*} e^{-2\pi i \gamma_0 \cdot p} \tilde{f}(p) \quad (\text{D.5})$$

where $\tilde{f}(p)$ is the Fourier transformation of $f(x)$, i.e.,

$$\tilde{f}(p) = \int_{\mathbb{R}^d} e^{2\pi i x \cdot p} f(x) d^d x \quad (\text{D.6})$$

Note that the integral is done over the entire Euclidean space.

The proof is outlined below. We introduce an auxiliary function

$$F(z) = \sum_{\gamma \in \Gamma} f(\gamma + z) \quad (\text{D.7})$$

which is, by construction, periodic on the lattice Γ and can be Fourier expanded

$$F(z) = \sum_{p \in \Gamma^*} e^{-2\pi i z \cdot p} \tilde{F}(p), \quad \tilde{F}(p) = \frac{1}{\text{vol}(\Gamma)} \int_{\Gamma_0} e^{2\pi i y \cdot p} F(y) d^d y \quad (\text{D.8})$$

One can plug the definition of F into its Fourier expansion and have

$$F(z) = \sum_{p \in \Gamma^*} e^{-2\pi i z \cdot p} \frac{1}{\text{vol}(\Gamma)} \int_{\Gamma_0} e^{2\pi i y \cdot p} \sum_{\gamma \in \Gamma} f(\gamma + y) d^d y$$

Then we can combine the summation of x and the integral of y into an integral over the entire space. Note that the unit cell Γ_0 is not shifted by γ_0 , we have

$$\begin{aligned} F(z) &= \sum_{p \in \Gamma^*} e^{-2\pi i z \cdot p} \frac{e^{-2\pi i \gamma_0 \cdot p}}{\text{vol}(\Gamma)} \int_{\mathbb{R}^d} e^{2\pi i x \cdot p} f(x) d^d x \\ &= \sum_{p \in \Gamma^*} e^{-2\pi i z \cdot p} \frac{e^{-2\pi i \gamma_0 \cdot p}}{\text{vol}(\Gamma)} \tilde{f}(p) \end{aligned}$$

Finally, setting $z = 0$ yields the Poisson summation formula.

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